Are passkeys as secure as you think?

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Intro – Fabian Bader

- Lives in Germany
- Cyber Security Architect @ glueck⬓kanja
- Microsoft MVP (Security / Azure)
- Organizer of "Purple Elbe Security User Group"
- Author of
  - TokenTacticsV2
  - entrascopes.com
  - XDRInternals
  - SentinelARConverter

Socials

Blog/talks:	cloudbrothers.info
Twitter:	@fabian_bader
BlueSky:	@fabian.bader.cloud

Talk Agenda

- What is a passkey?
- Synced vs. Device-bound passkeys
- Security of passkey providers
- Why does attestation matter?
- Threat modeling for enterprises & Currently known attacks
- Attack mitigation

What is a passkey?

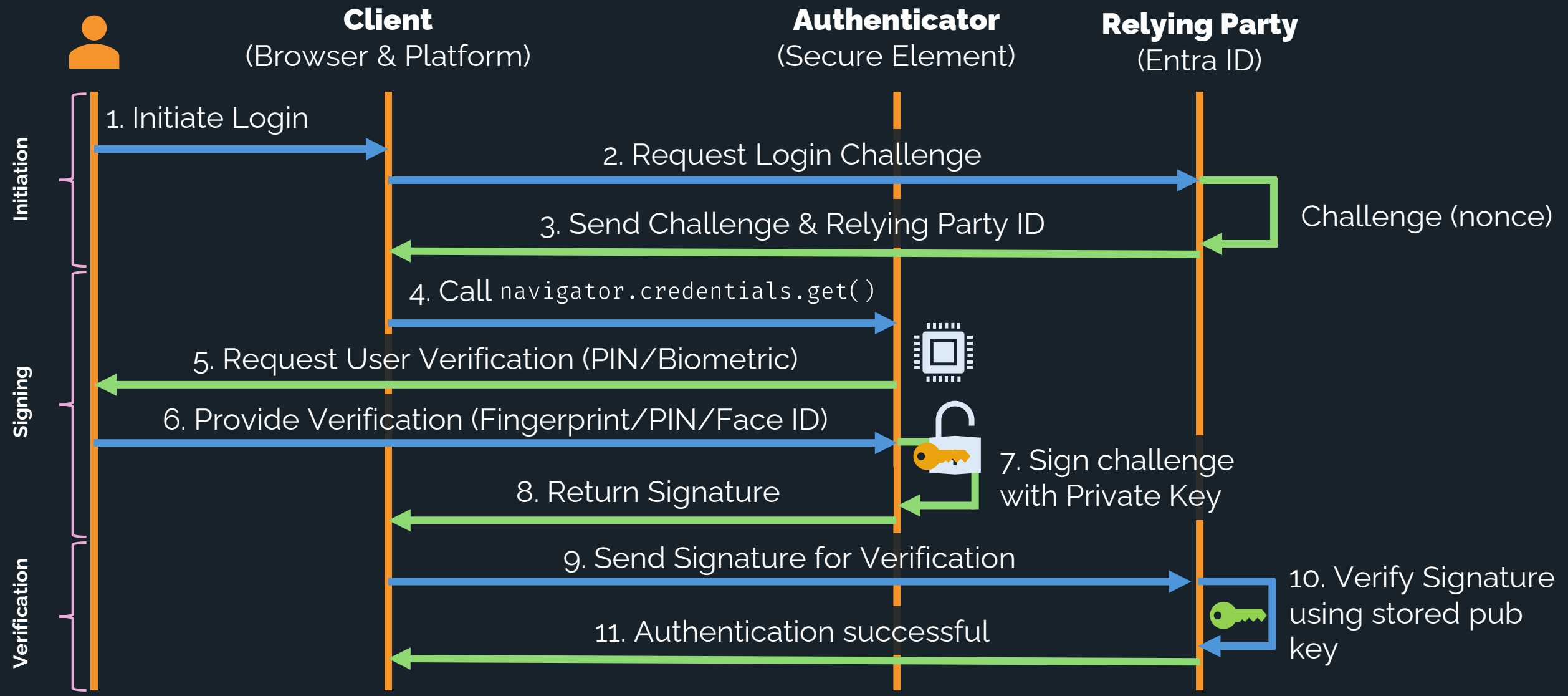
What is a passkey?

- FIDO2/WebAuthn Discoverable Credential
  - DC: Stores Relying Party Id & User ID

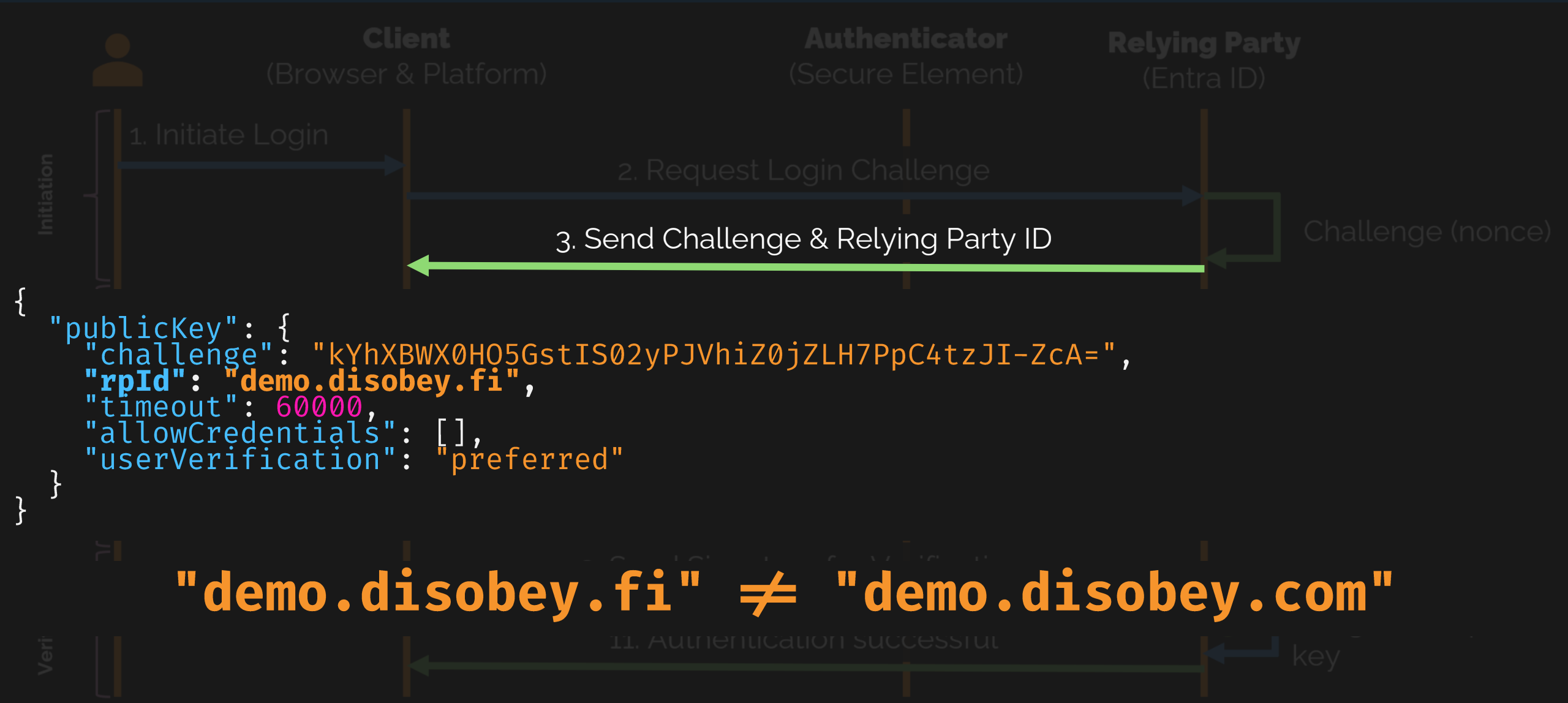
replyingPartyId	Username
login.microsoft.com	maija@c4a8korriban.com

- Based on a cryptographic foundation
  - Private key stored in the authenticator
  - Public key stored by the relying party
- Enforced origin binding

Authentication with a passkey



Why phishing resistant?

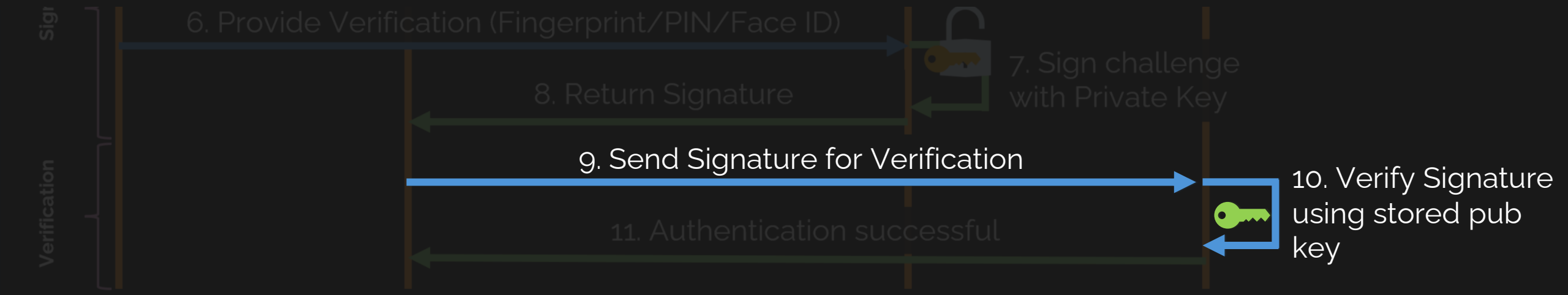



```
{
  "publicKey": {
    "challenge": "kYhXBWX0HO5GstIS02yPJVhiZ0jZLH7PpC4tzJI-ZcA=",
    "rpId": "demo.disobey.fi",
    "timeout": 60000,
    "allowCredentials": [],
    "userVerification": "preferred"
  }
}
```

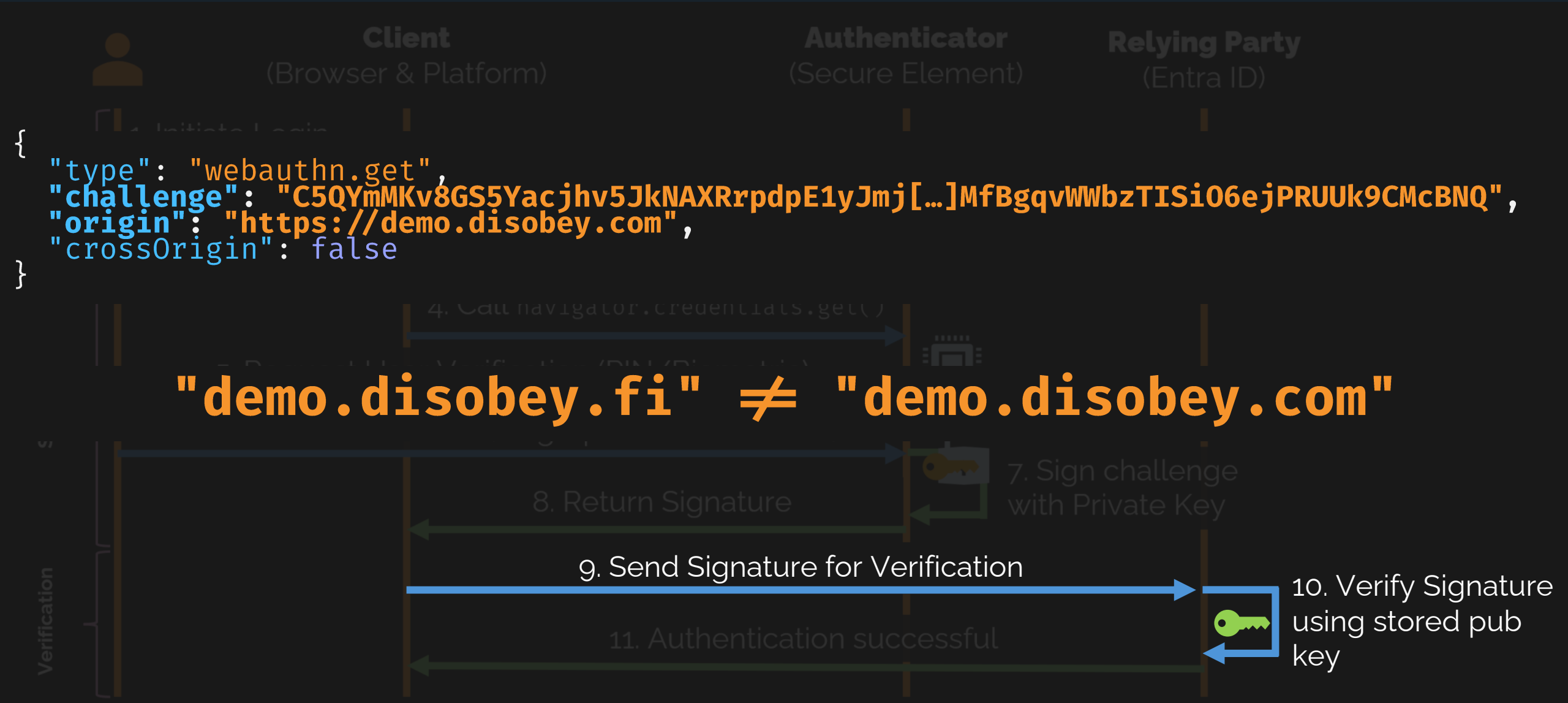

"demo.disobey.fi" ≠ "demo.disobey.com"

Why phishing resistant?


```
{
  "id": "0PNe7hW8UGhxeEfagFJuq6L6KkGhjli5JuGnOJu07Ak",
  "rawId": "0PNe7hW8UGhxeEfagFJuq6L6KkGhjli5JuGnOJu07Ak",
  "response": {
    "authenticatorData": "dKbqkhPJnC90siSSsyDPQCYqlMGpUKA5fyklC2CEHvAFAAAAAg",
    "clientDataJSON": "eyJ0eXBlIjoid2ViYXV0aG4uZ2[…]IiwiY3Jvc3NPcmlnaW4iOmZhbHNlfQ",
    "signature": "MEUCIQDOBnmbmDAIN37cSQYX5QrpnDZB […]YADsog_KaY3CbL2FCQ",
    "userHandle": "d2ViYXV0aG5pby10ZXN0QHRlc3QuY28"
  },
  "type": "public-key",
  "clientExtensionResults": {},
  "authenticatorAttachment": "platform"
}
```

Why phishing resistant?

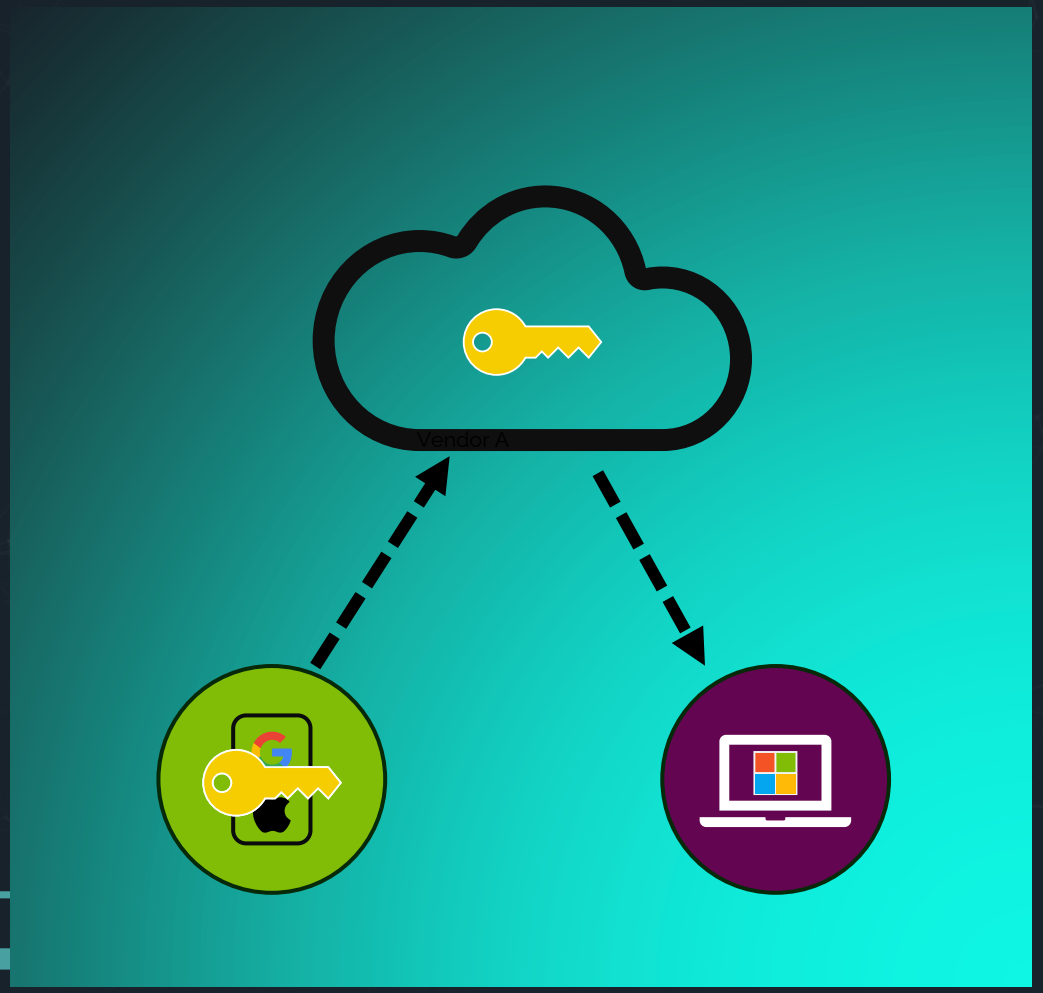



```
{
  "type": "webauthn.get",
  "challenge": "C5QYmMKv8GS5Yacjhv5JkNAXRrpdpE1yJmj[…]MfBgqvWWbzTISiO6ejPRUUk9CMcBNQ",
  "origin": "https://demo.disobey.com",
  "crossOrigin": false
}
```


"demo.disobey.fi" ≠ "demo.disobey.com"

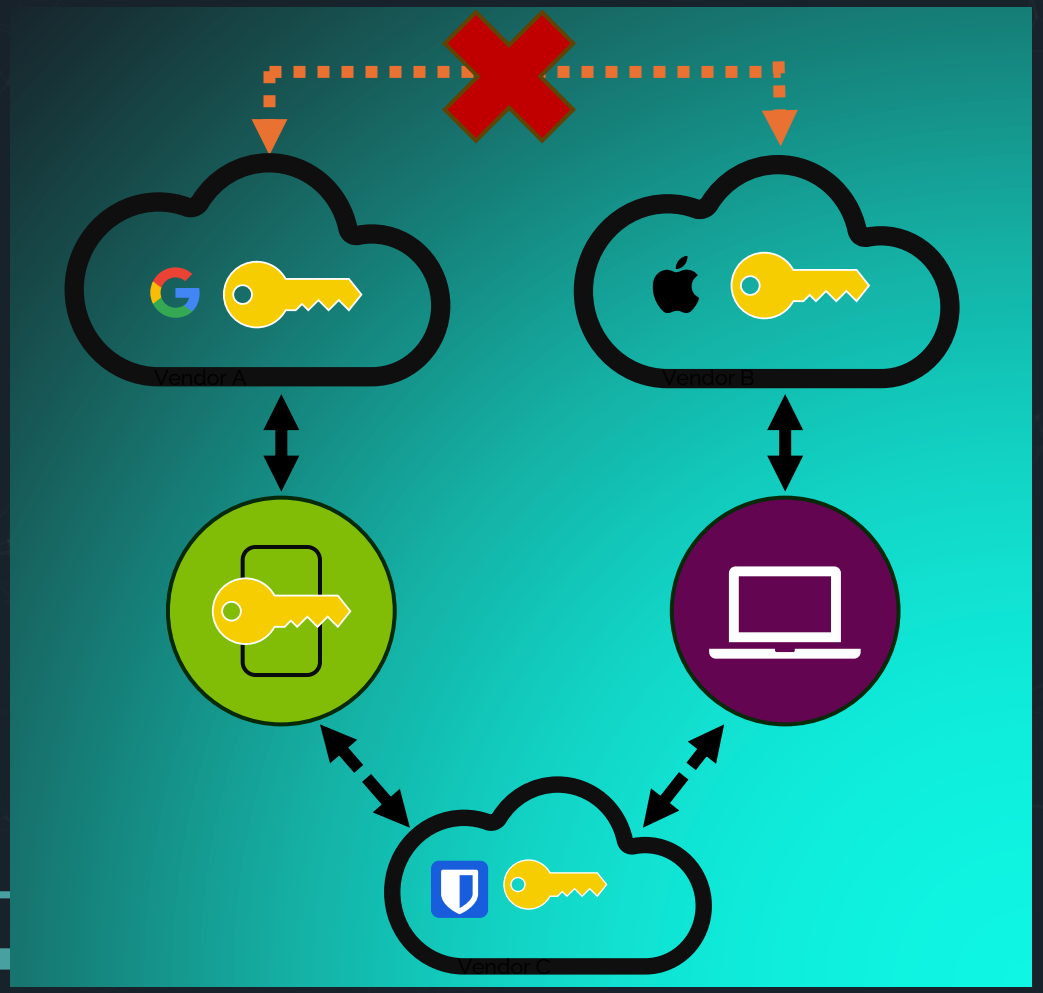
Synced vs. device-bound passkeys

Synced vs. Device-bound passkeys



- Passkeys are synced by **default**
- Private key is sent to mobile device vendor or third-party passkey provider
- Restore and usage on other devices possible

Synced vs. Device-bound passkeys

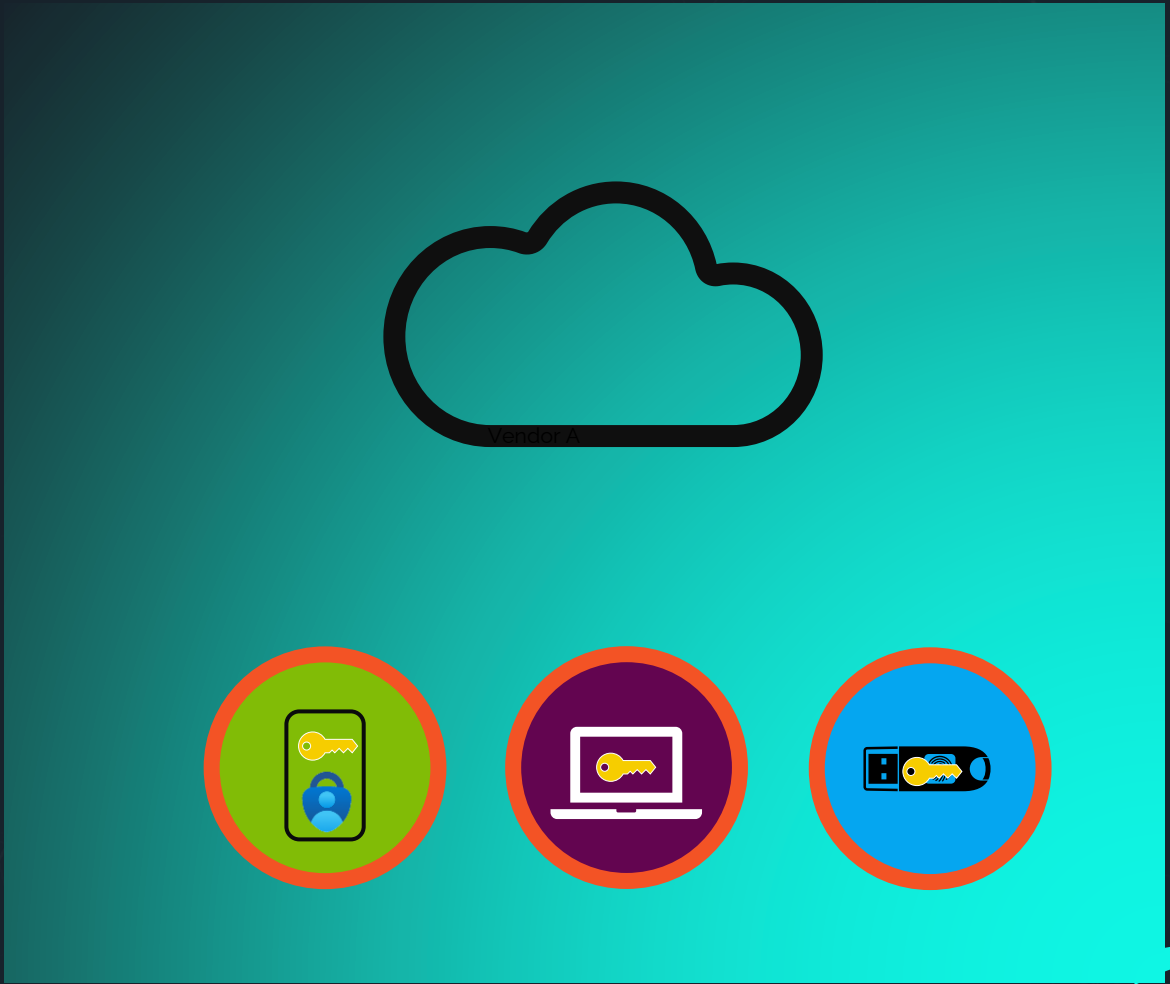


- Native cross vendor sync is not (yet) possible
- Workarounds
  - Cross-Device Authentication
  - Third-party passkey provider
  - Export/Import (e.g. JSON)
- Credential eXchange Protocol
  - Support on Apple devices

Synced vs. Device-bound passkeys

- The private key cannot leave the device
- FIDO2 security keys are device-bound passkeys
- Some apps create a device-bound passkey (e.g. Microsoft Authenticator)

- Recovery = New Setup



Does attestation matter?

Attestation

- Each make and model is identified by an AAGUID
  - 128-bit identifier (UUID)
  - YubiKey 5C Firmware 5.1 = cb69481e-8ff7-4039-93ec-0a2729a154a8
  - YubiKey 5C Firmware 5.7 = 19083c3d-8383-4b18-bc03-8f1c9ab2fd1b
- Burnt into the device by the manufacturer
- Small impact to user privacy
- Synced passkeys do not support attestation*
- Enterprise attestation allows to identify certain FIDO2 keys instead of just a make and model

*Attestation is possible but not implement by any major Credential Manager

Attestation

MDS Browser e.g. https://aaguid.nicolasuter.ch/

AAGUIDs

aaguid.nicolasuter.ch

AAGUIDs

AAGUIDs are unique identifiers for FIDO2 authenticators such as passkeys and physical keys and are used for attestation purposes.
This list is based on the FIDO Alliance Metadata Service.

Display unofficial passkey blob

2025-12-14 Last Update

2026-01-01 Next Update

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Name	AAGUID	Authn Algorithms	Key Protection	Attachment Hint
ACS FIDO Authenticator	50a45b0c-80e7-f944-bf29-f552bfa2e048	secp256r1_ecdsa_sha256_raw	hardware,secure_element	external,wired
ACS FIDO Authenticator Card	973446ca-e21c-9a9b-99f5-9b985a67af0f	secp256r1_ecdsa_sha256_raw	hardware,secure_element	external,wireless,nfc
ACS FIDO Authenticator NFC	c89e6a38-6c00-5426-5aa5-c9cbf48f0382	secp256r1_ecdsa_sha256_raw	hardware,secure_element	external,wired,wireless,nfc
Allthenticator Android App: roaming BLE FIDO2 Allthenticator for Windo...	5ca1ab1e-fa57-1337-f1d0-a117371ca702	rsassa_pkcsv15_sha256_raw,secp256r1_ecdsa_sha256_raw	hardware,secure_element	external,wired
Allthenticator iOS App: roaming BLE FIDO2 Allthenticator for Windows, ...	5ca1ab1e-1337-fa57-f1d0-a117e71ca702	secp256r1_ecdsa_sha256_raw	hardware,secure_element	external,wired
Android Authenticator	b93fd961-f2e6-462f-b122-82002247de78	secp256r1_ecdsa_sha256_raw	hardware,tee	internal
Arculus FIDO 2.1 Key Card [P71]	3f59672f-20aa-4afe-b6f4-7e5e916b6d98	secp256r1_ecdsa_sha256_raw	hardware,secure_element	external,wired
Arculus FIDO2/U2F Key Card	9d3df6ba-282f-11ed-a261-0242ac120002	secp256r1_ecdsa_sha256_raw	hardware,secure_element	external,wired
ATKey.Card CTAP2.0	d41f5a69-b817-4144-a13c-9ebd6d9254d6	secp256r1_ecdsa_sha256_raw	hardware,secure_element	external
ATKey.Card NFC	da1fa263-8b25-42b6-a820-c0036f21ba7f	secp256r1_ecdsa_sha256_raw	hardware,secure_element	external,wired,wireless,nfc
ATKey.Pro CTAP2.0	e1a96183-5016-4f24-b55b-e3ae23614cc6	ed25519_eddsa_sha512_raw,secp256r1_ecdsa_sha256_raw	hardware	external

Passkeys = Easily faked AAGUID

```qt/qmake
32 #include <botan/pkcs8.h>
33 #include <botan/pubkey.h>
34 #include <botan/rsa.h>
35 #include <botan/sodium.h>
36
37 #include <bitset>
38
39 Q_GLOBAL_STATIC(BrowserPasskeys, s_browserPasskeys);
40
41 // KeePassXC AAGUID: fdb141b2-5d84-443e-8a35-4698c205a502
42 const QString BrowserPasskeys::AAGUID = QStringLiteral("fdb141b25d84443e8a354698c205a502");
43
44 // Authenticator capabilities
45 const QString BrowserPasskeys::ATTACHMENT_CROSS_PLATFORM = QStringLiteral("cross-platform");
46 const QString BrowserPasskeys::ATTACHMENT_PLATFORM = QStringLiteral("platform");
47 const QString BrowserPasskeys::AUTHENTICATOR_TRANSPORT_INTERNAL = QStringLiteral("internal");
48 const QString BrowserPasskeys::AUTHENTICATOR_TRANSPORT_NFC = QStringLiteral("nfc");
49 const QString BrowserPasskeys::AUTHENTICATOR_TRANSPORT_USB = QStringLiteral("usb");
50 const bool BrowserPasskeys::SUPPORT_RESIDENT_KEYS = true;
51 const bool BrowserPasskeys::SUPPORT_USER_VERIFICATION = true;
52
53 const QString BrowserPasskeys::PUBLIC_KEY = QStringLiteral("public-key");
54 const QString BrowserPasskeys::REQUIREMENT_DISCOURAGED = QStringLiteral("discouraged");
55 const QString BrowserPasskeys::REQUIREMENT_PREFERRED = QStringLiteral("preferred");
56 const QString BrowserPasskeys::REQUIREMENT_REQUIRED = QStringLiteral("required");
```

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32 #include <botan/pkcs8.h>
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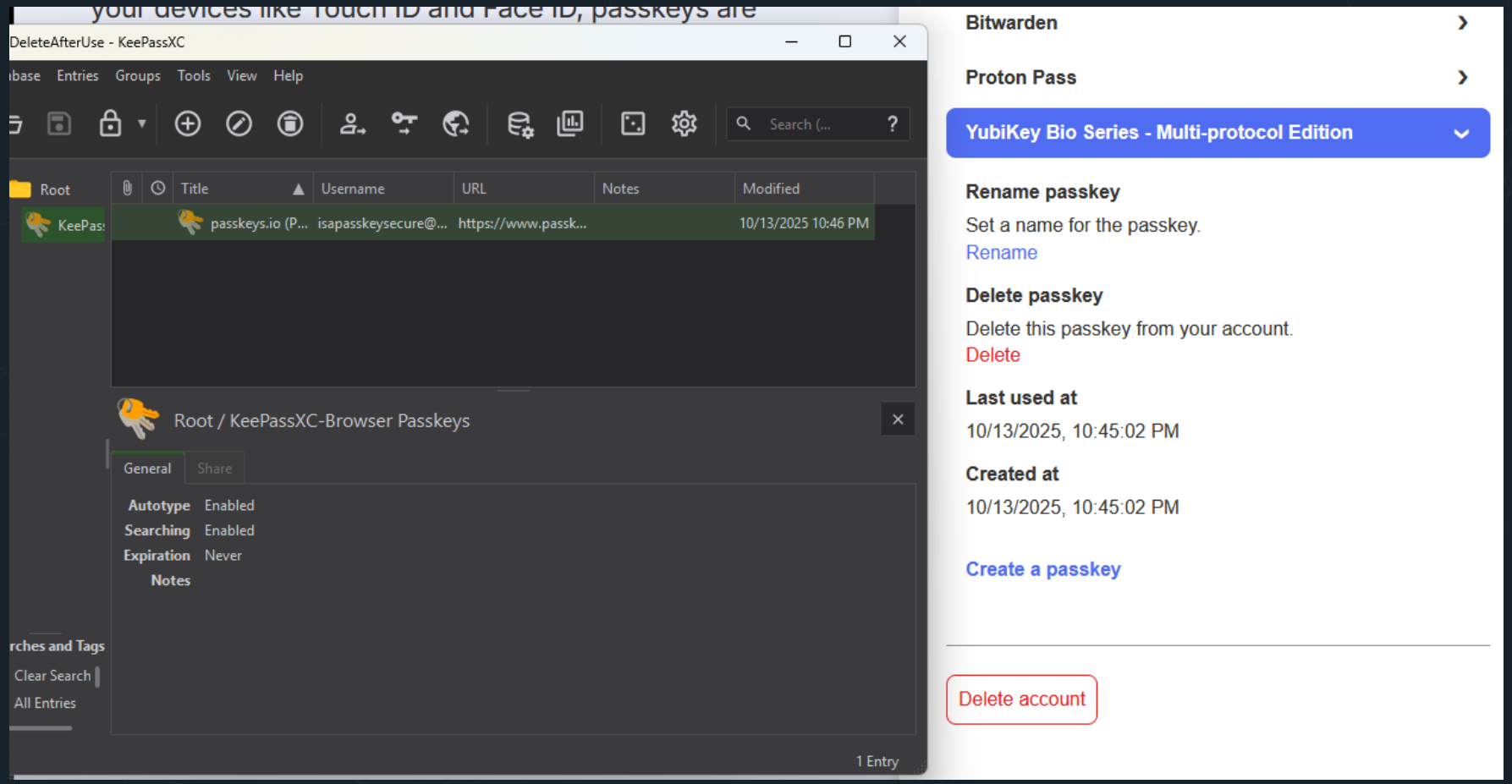
Passkeys = Easily faked AAGUID



Attestation

1. Credential key pair generated
2. Sign public key with private attestation key

3. Send signed public key to Relying Party

AAGUID: dd86a2da-86a0-4cbe-b462-4bd31f57bc6f
Vendor: Yubikey
Product: YubiKey Bio - FIDO Edition
Firmware: 5.7

4. Request Certificate information from MDS
5. Validate signed public key
6. Store public key with user object

FIDO MDS

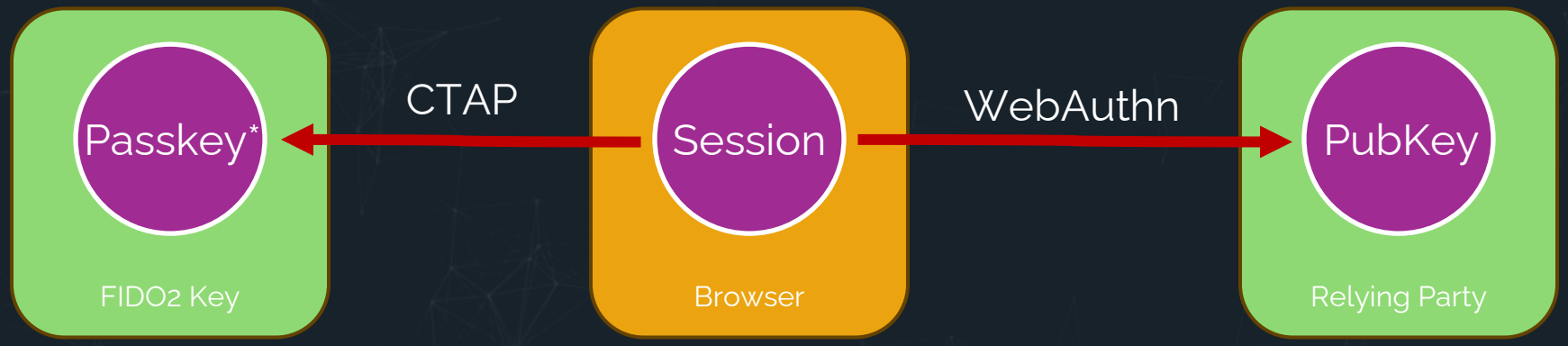
https://fidoalliance.org/fido-technotes-the-truth-about-attestation/

Syncable Passkeys do not support attestation*

*Attestation is possible but not implement by any major Credential Manager

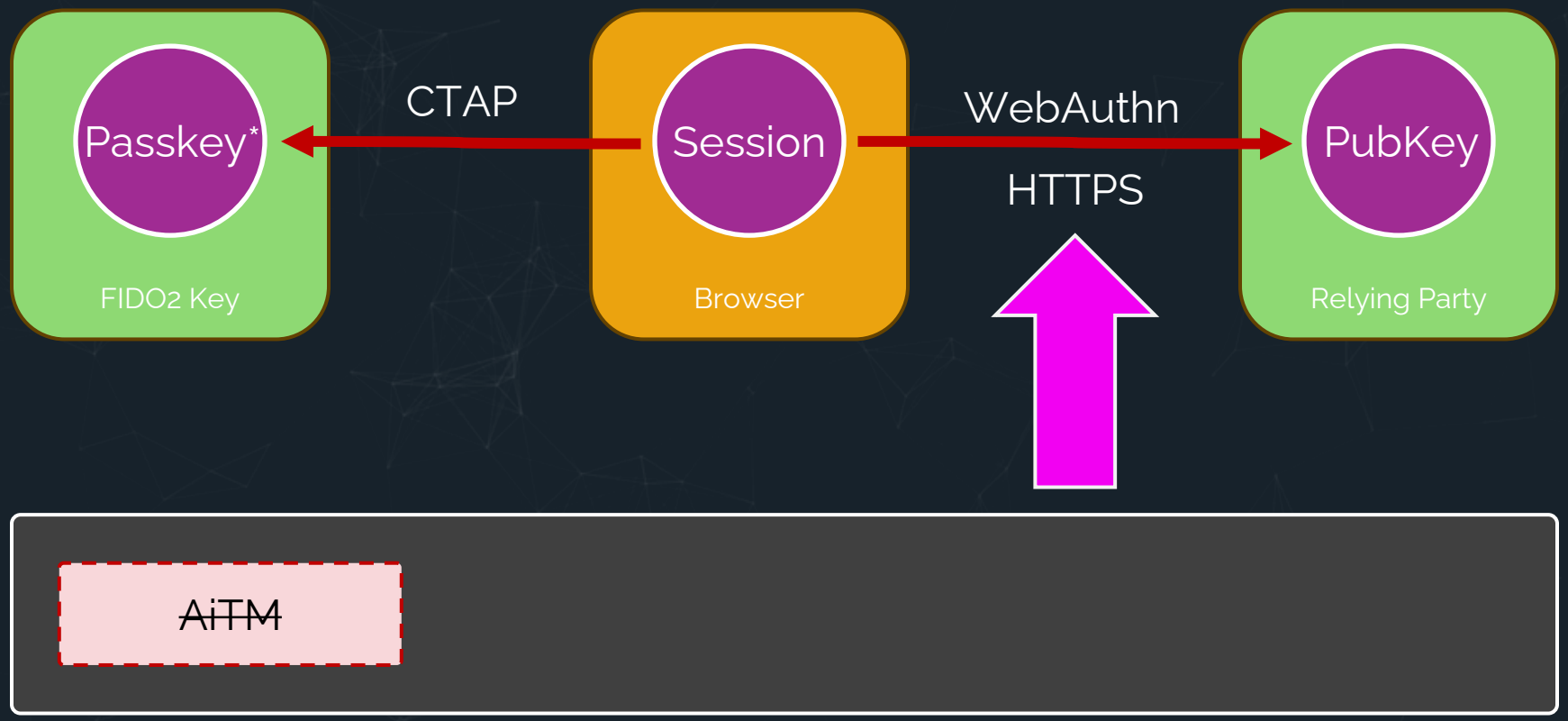
Threat modeling for enterprises

Attack vectors: Device-bound passkeys



*Passkey = Private Key

Attack vectors:
Device-bound passkeys

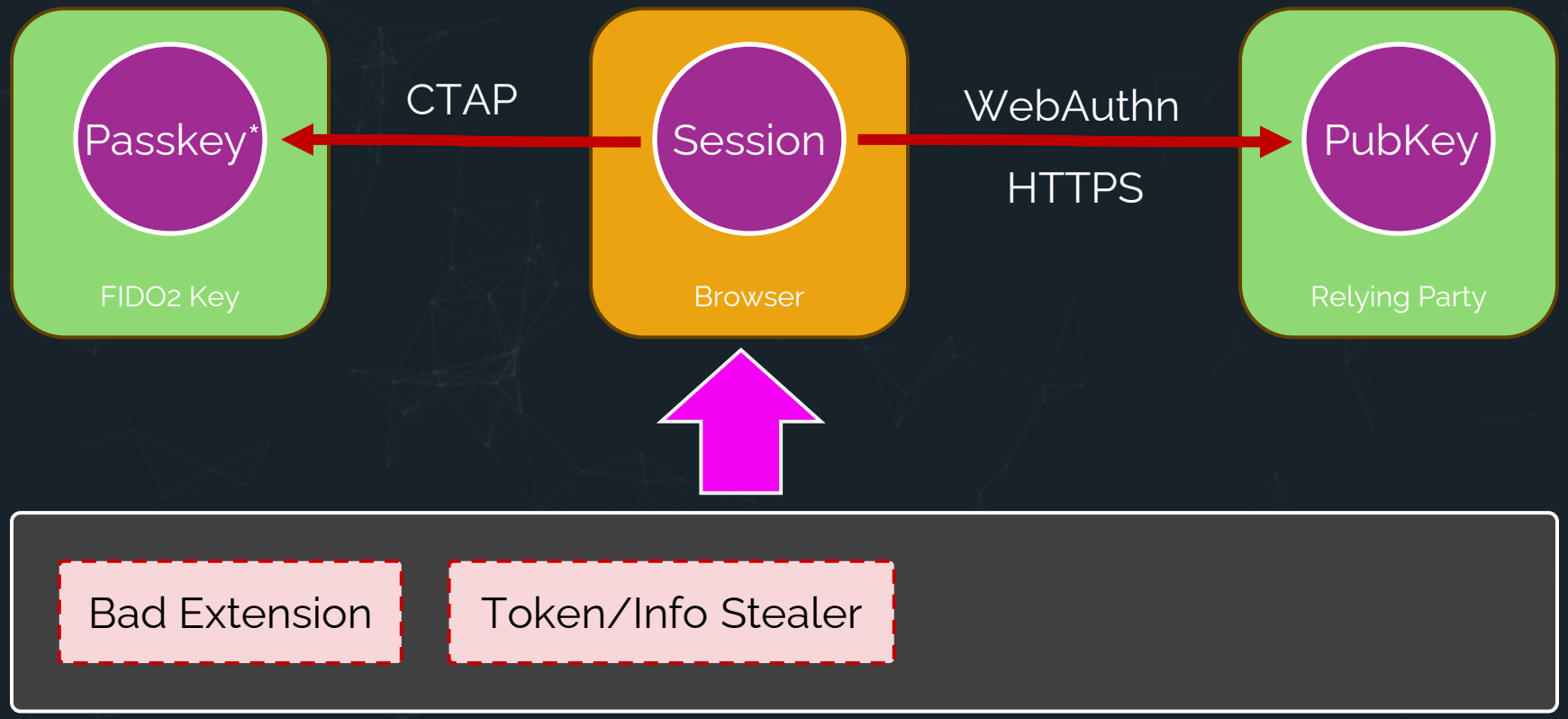


*Passkey = Private Key

Downgrade attacks

- AiTM "hide" the Passkey sign-in option from the victim
- If already selected the AiTM plugin will force a fallback to another, phishable method
- Security of the Passkey credential is not impacted
- Not a "real" attack on passkey, but still relevant

Attack vectors: Device-bound passkeys



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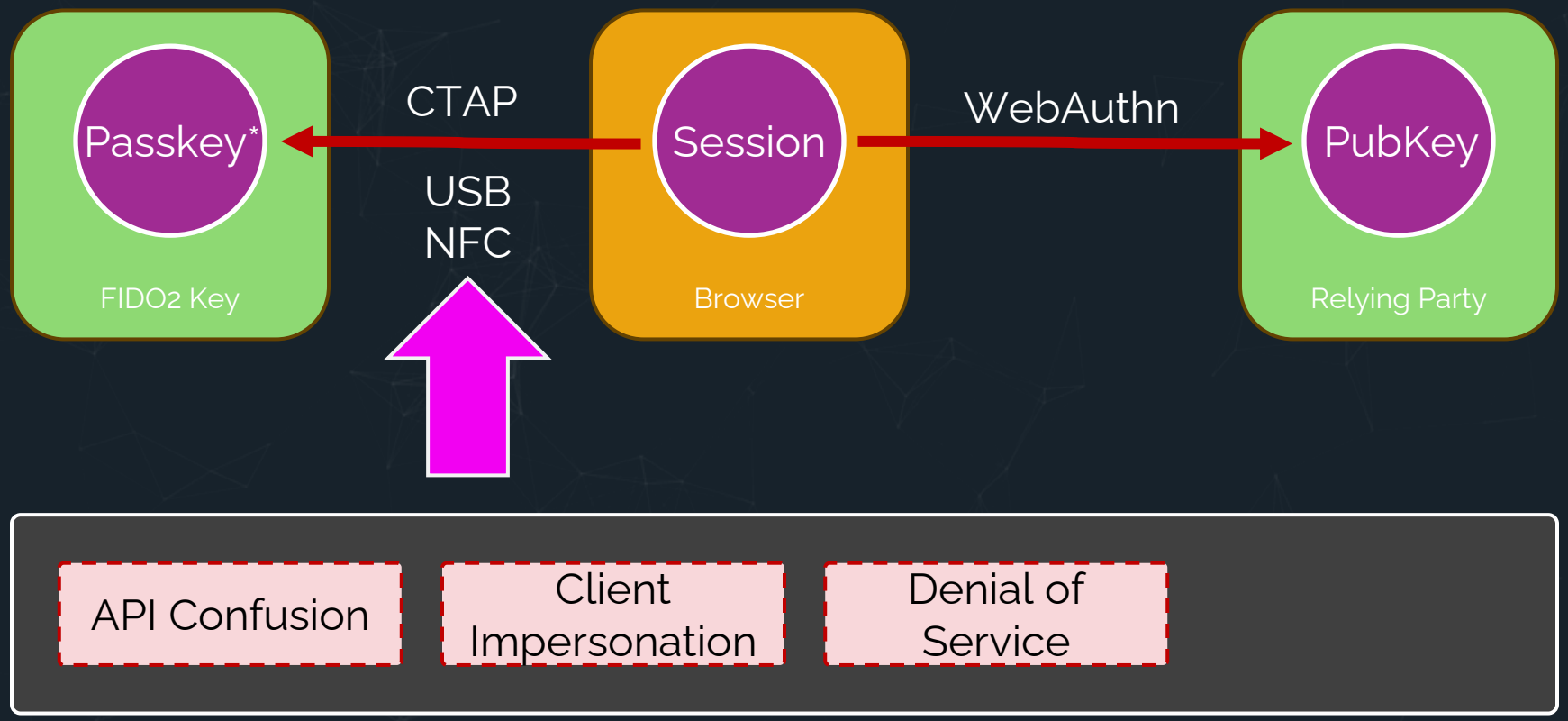
Attack vectors: Device-bound passkeys

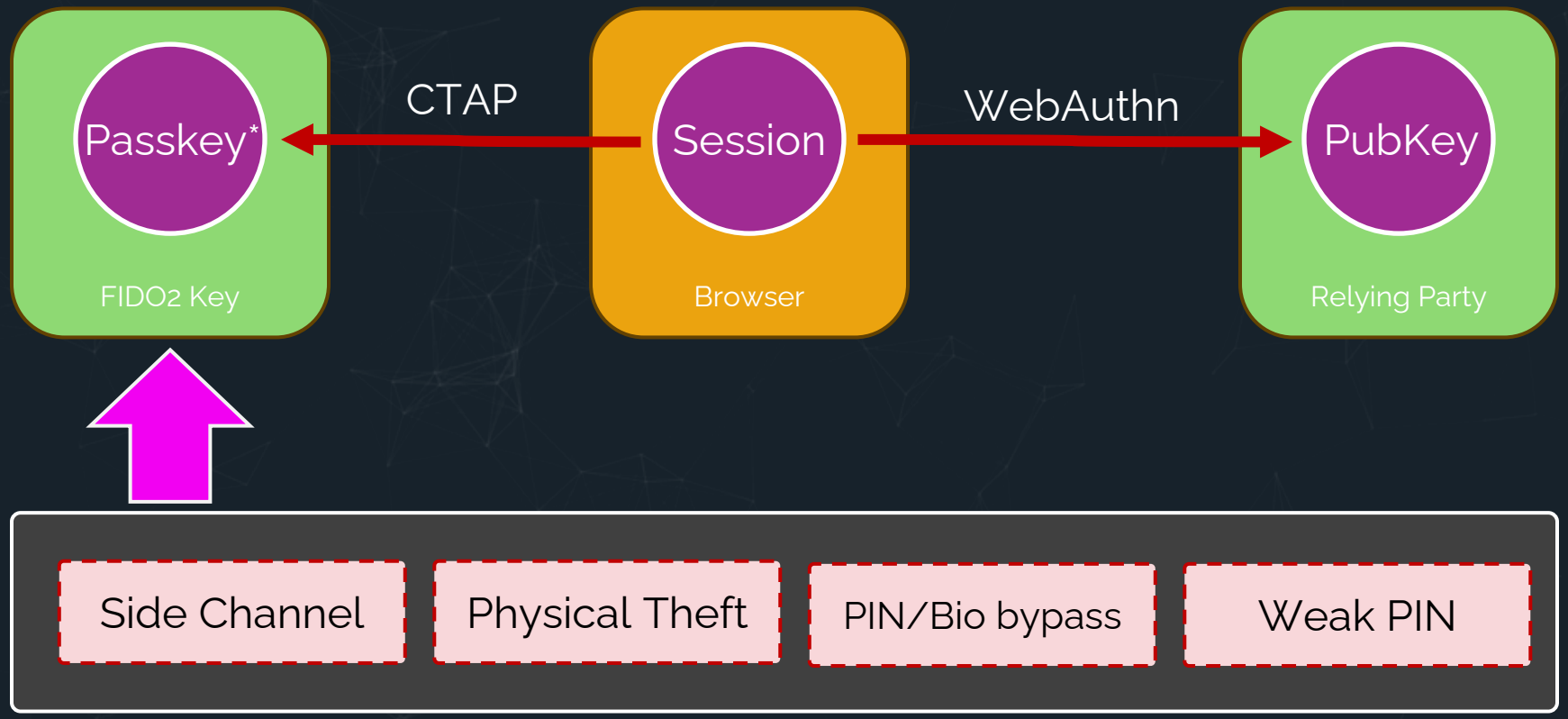


*Passkey = Private Key

Known attacks - CTRAPS

- CTRAPS: CTAP Client Impersonation and API Confusion on FIDO2
  - Marco Casagrande, EURECOM & Daniele Antonioli, EURECOM
- Attacks
  - Delete Credentials (NFC)
  - List Credentials
- Sources:
  - https://arxiv.org/pdf/2412.02349
  - https://www.youtube.com/watch?v=07B0etOq7OM

Attack vectors: Device-bound passkeys

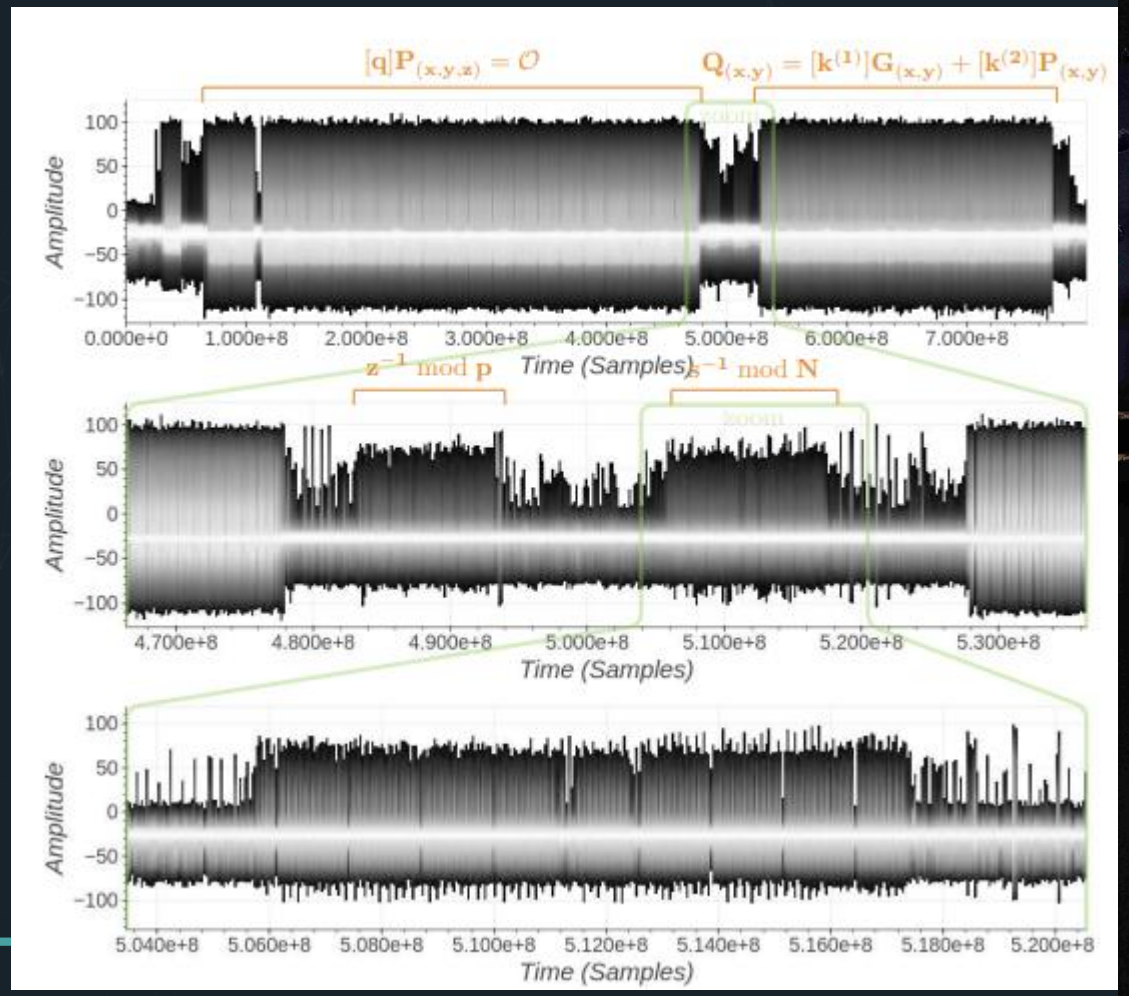


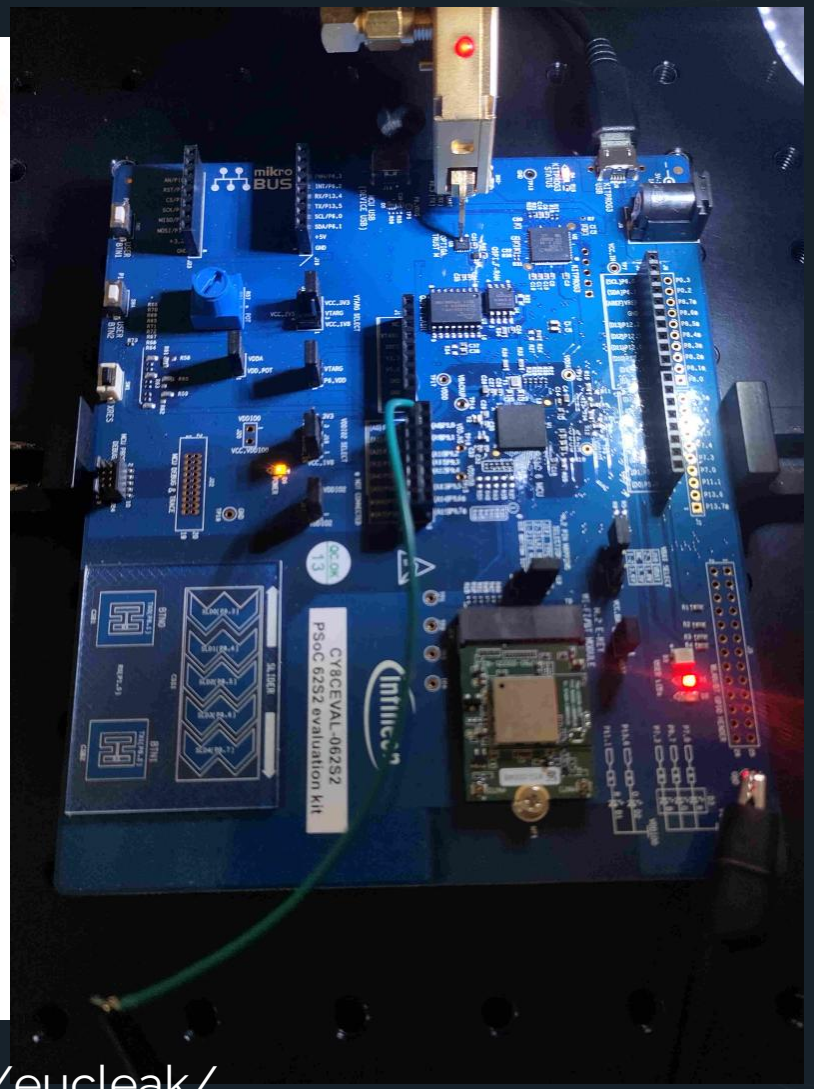
*Passkey = Private Key

Known attacks - CVE-2024-45678

- Side-Channel attack on Infineon SLE78
- Only when used with Infineon cryptographic library
- Allows for extraction of private keys
- Affected: All YubiKey 5 Series < 5.7
- Requirements:
  - Physical access
  - PIN for the device
- Research published: https://ninjalab.io/eucleak/

Known attacks - CVE-2024-45678





Source: https://ninjalab.io/eucleak/

Known attacks - CVE-2024-45678

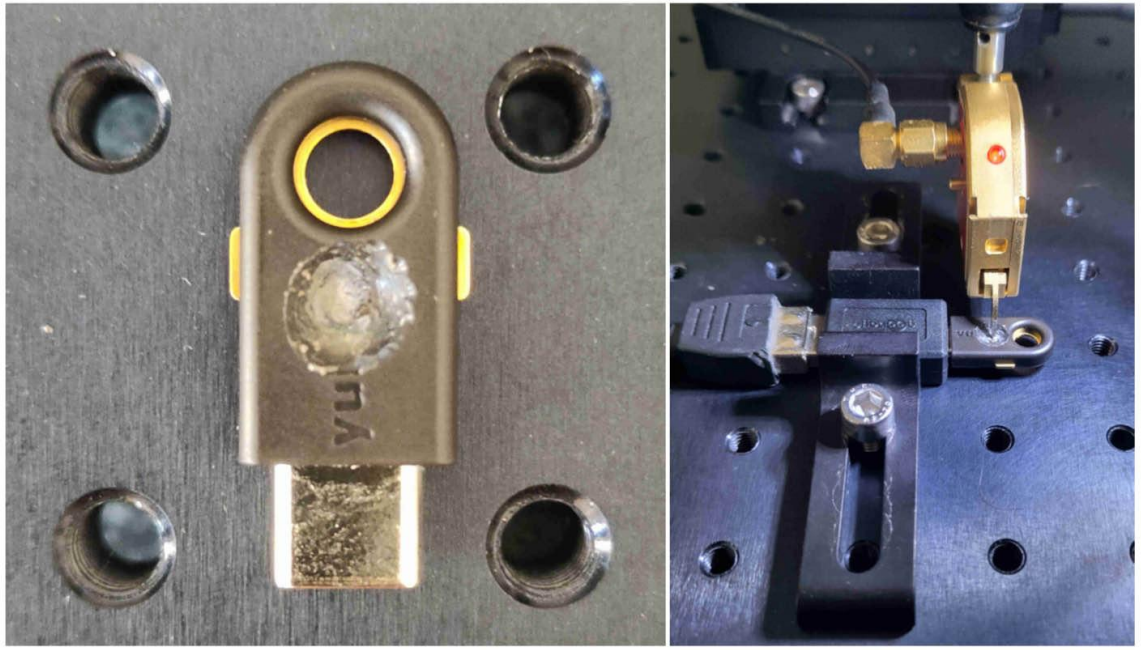


Figure A.3: `YubiKey 5C` – Second Opening

In both cases however, the device needs to be re-packaged if the adversary wants to give it back to legitimate user without him noticing. We did not study further this issue.

Source: https://ninjalab.io/eucleak/

Impact on attestation

Attestation

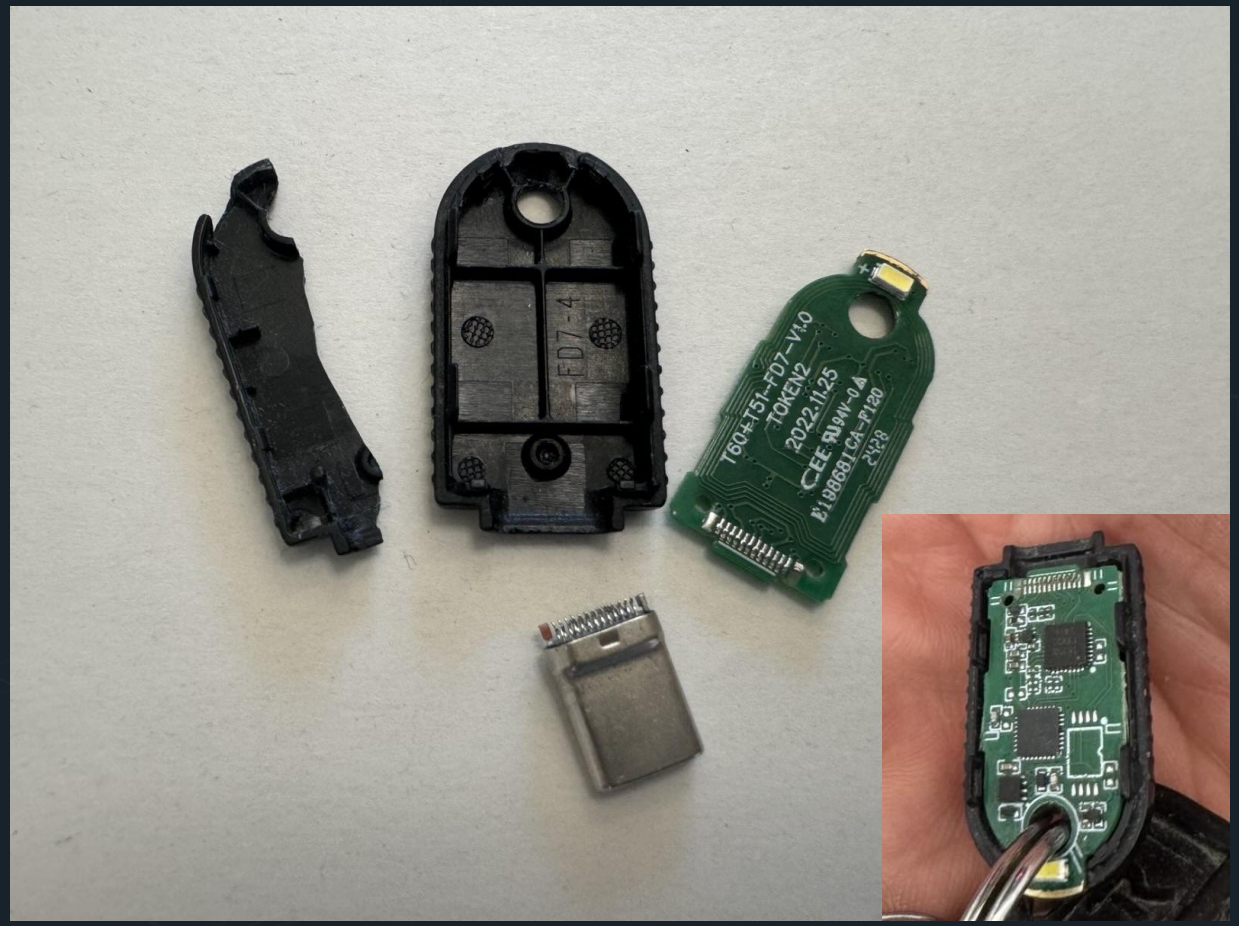
Attestation is built-in to the FIDO and WebAuthn protocols. This feature enables each relying party to use a cryptographically verified chain of trust from the device's manufacturer to choose which security keys to trust. This feature is shown as allow lists and disallow lists of AAGUIDs in an identity provider that may be customizable for organizations.

An attacker could exploit this issue to create a fraudulent YubiKey using the recovered attestation key. This would produce a valid FIDO attestation statement during the make credential resulting in a bypass of an organization's authenticator model preference controls for affected YubiKey versions.

Organizations relying on FIDO attestation to ensure genuine YubiKeys are in use may consider supplementing FIDO login with other credentials such as YubiOTP or RSA attestation statements from PIV or OpenPGP. For more information about FIDO attestation and detailed instructions, see the related support article.

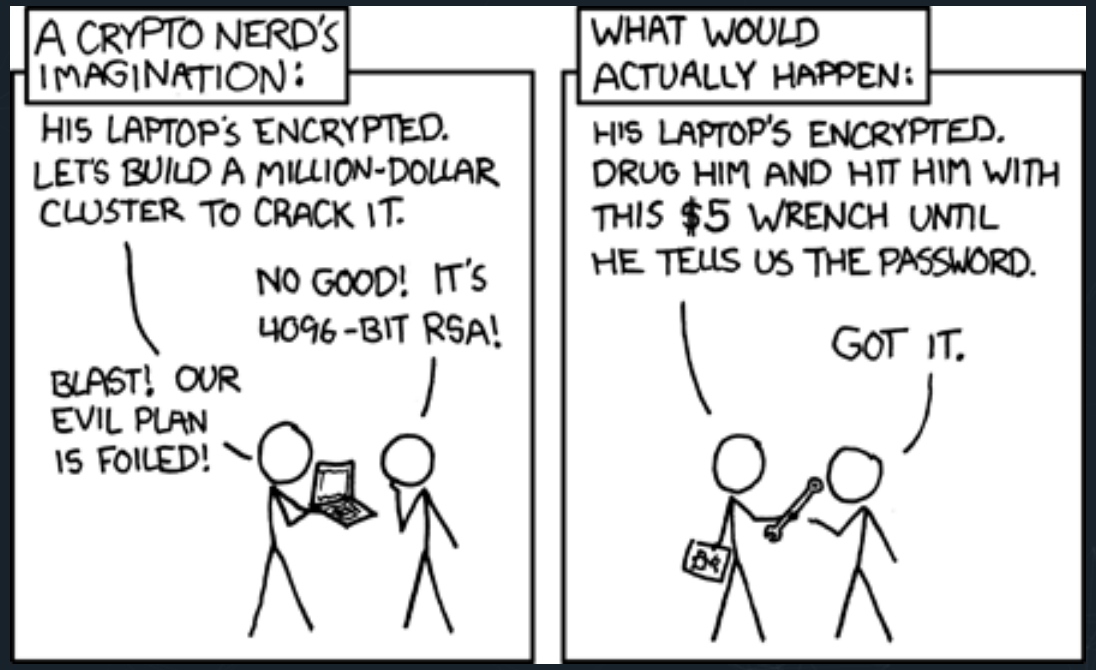
Source:
https://www.yubico.com/support/security-advisories/ysa-2024-03/

Other hardware token



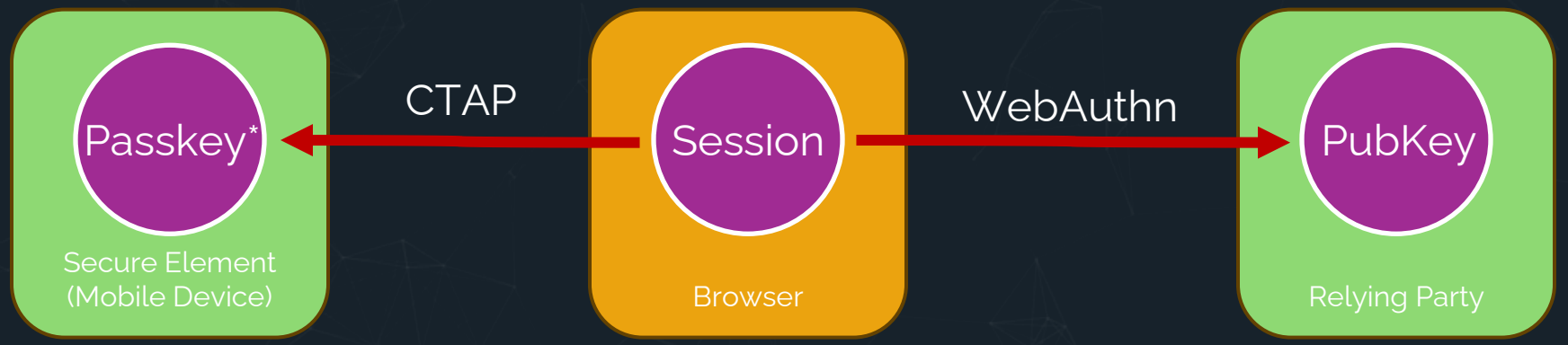
Photos by Jakob Schaefer

Easier method



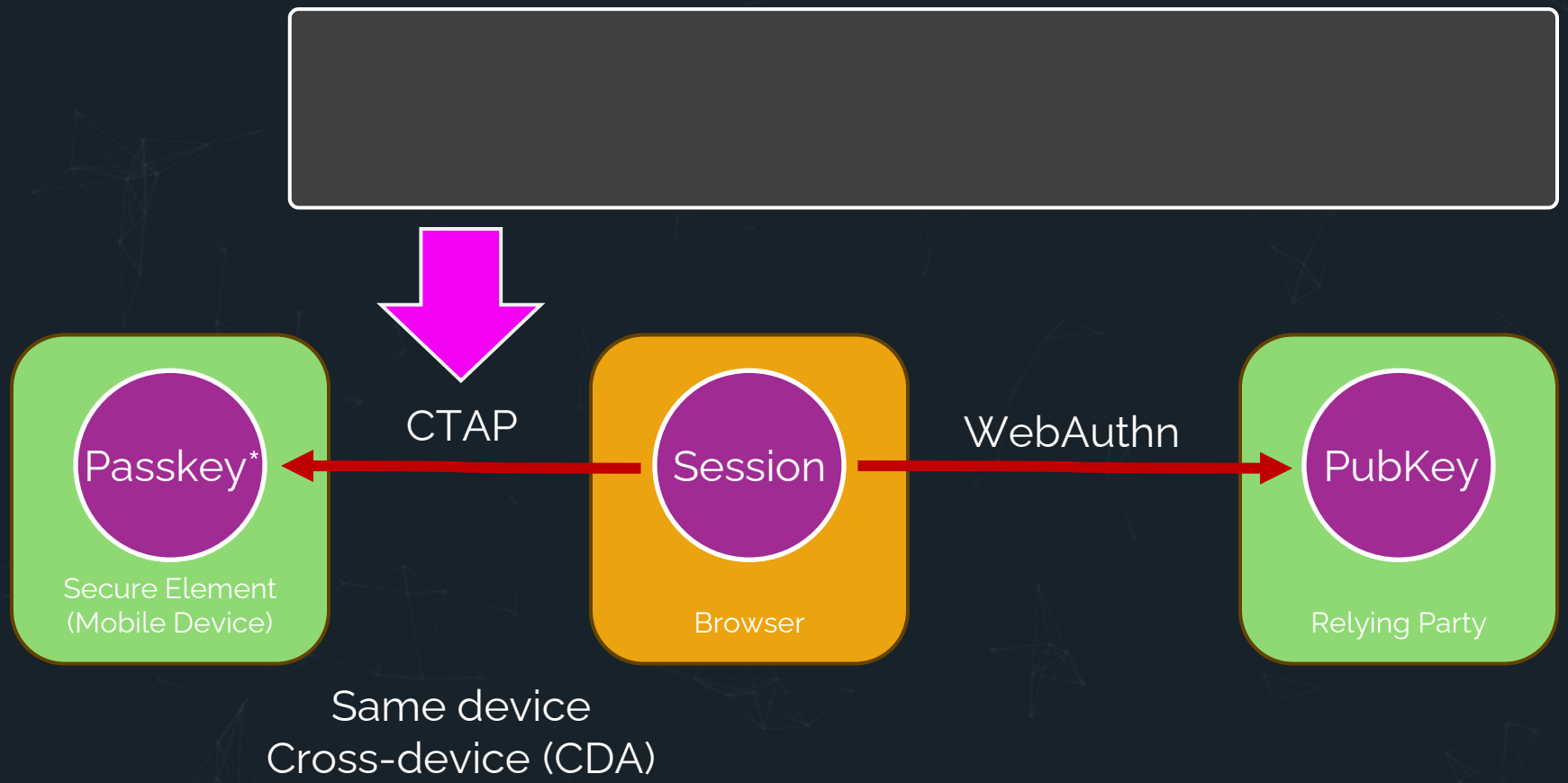
Source: https://xkcd.com/538/

Attack vectors: Syncable passkeys



*Passkey = Private Key

Attack vectors: Syncable passkeys



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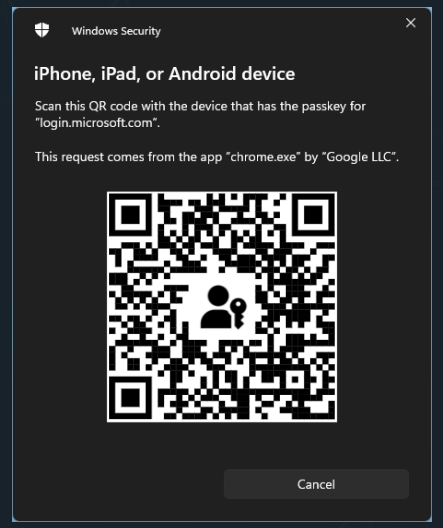
Cross Device Authentication

Cross Device Authentication



- Bluetooth (BLE) on both devices for proximity check
- Internet access for data transfer
  - https://cable.ua5v.com (Android)
  - https://cable.auth.com (Apple)

FIDO:/088521772645746
304256629196898023805
213791974887885159969
946751928771388701793
485401070923423366366
303159168738737767290
060661159865120837177

QR Code unraveled

```
FIDO:/088521772645746
304256629196898023805
213791974887885159969
946751928771388701793
485401070923423366366
303159168738737767290
060661159865120837177
011010667266107096654
083332
```

- Base10 encoded string
- Concise Binary Object Representation (CBOR) data format

QR Code unraveled

FIDO:/088521772645746
304256629196898023805
213791974887885159969
946751928771388701793
485401070923423366366
303159168738737767290
060661159865120837177
011010667266107096654
083332

- Base10 encoded string
- Concise Binary Object Representation (CBOR) data format

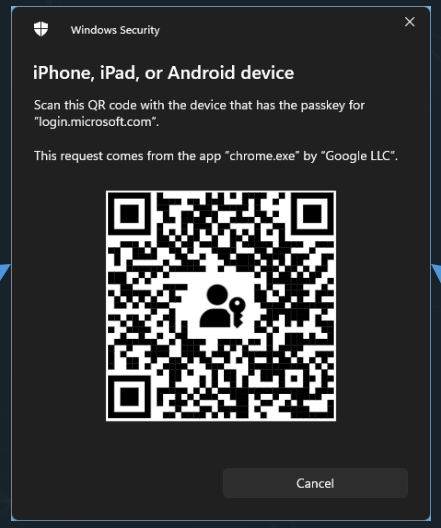
QR Code unraveled

```
A6 00 58 21 02 73 1F
00 0E 75 37 28 D1 39
97 00 CD 91 98 8A EA
85 12 00 2D B4 16 91
2E D5 38 00 7A 17 FF
52 2B 56 31 00 5F C4
01 50 7A F8 FB 00 59
60 2E FD 4F 8C 38 00
48 AF DA C4 B5 27 02
00 02 03 1A 67 4B 14
84 00 04 F5 05 62 67
61 00 00
```

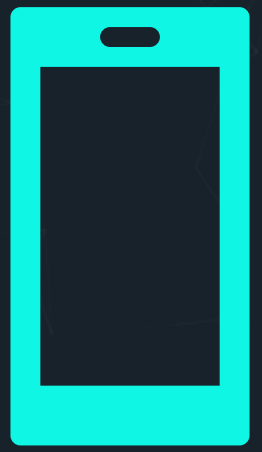
```
// Compressed public key
0: h'02731F0E7537[ ... ]315FC4‘
// Shared secret
1: h'7AF8FB59602E[ ... ]C4B527‘
// decodeTunnelServerDomain
2: 2
// Current epoch time
3: 1732973700
// State-assisted transactions
4: true
// getAssertion or makeCredential
5: "ga"
```

Cross Device Authentication

1. Visit website and choose passkey signin
2. Select Cross Device Authentication
3. Generate QR Code



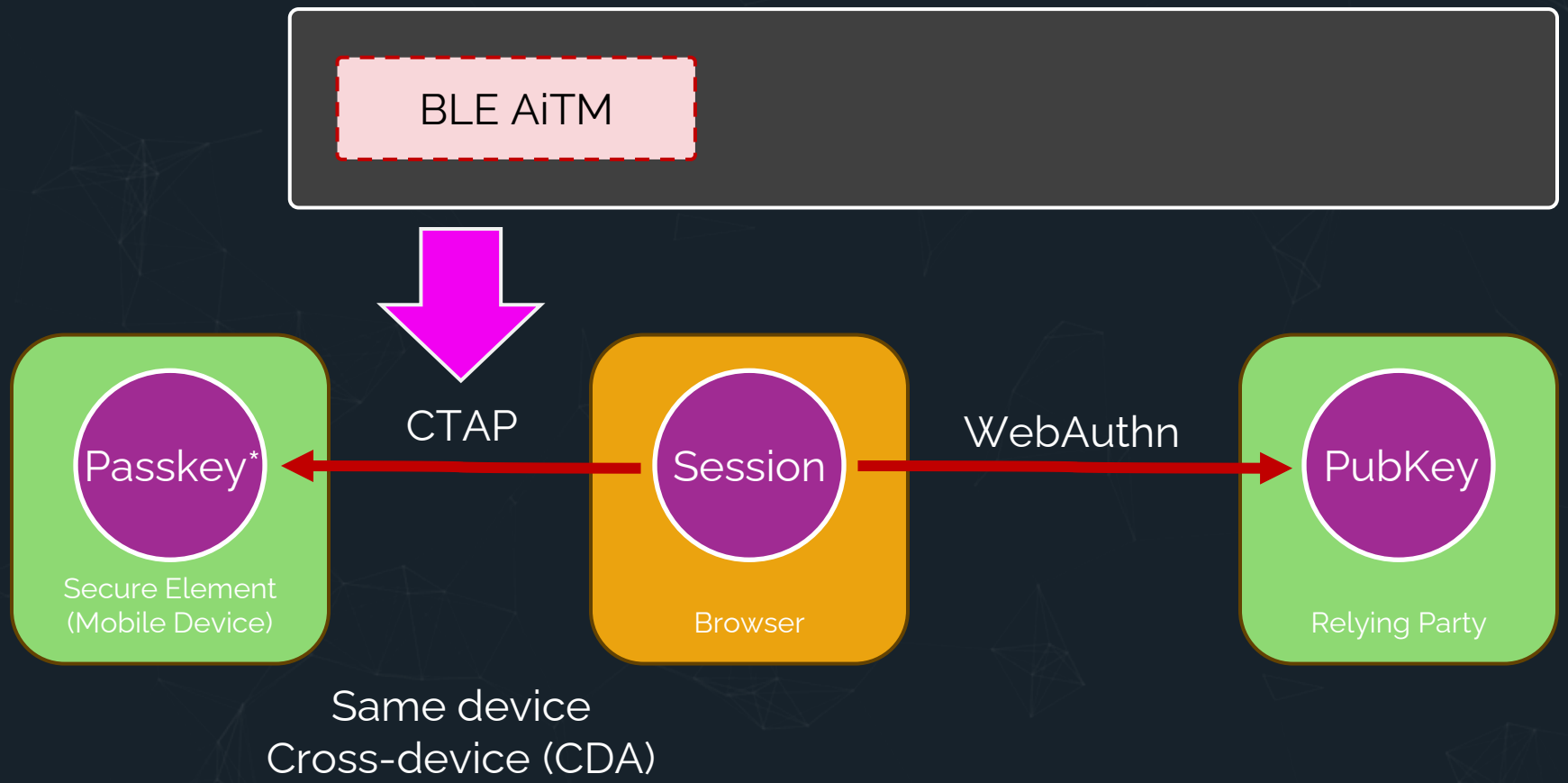
4. Scan QR Code and retrieve secret



5. Broadcast BLE encrypted with secret

6. Receive message and connect through tunnel service

Attack vectors: Syncable passkeys

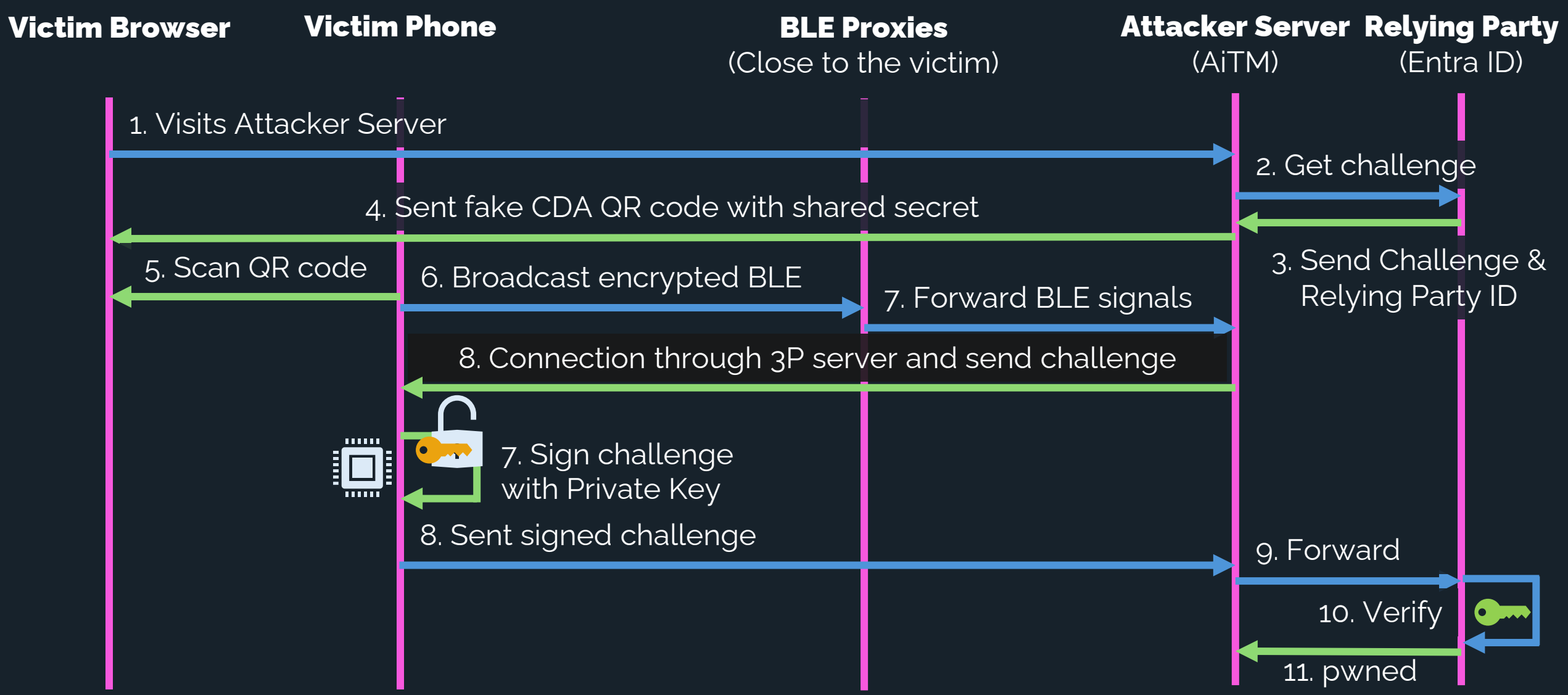


*Passkey = Private Key

Known attacks

- BLE AiTM aka Cross Device Phishing
- Sources:
  - https://mastersplinter.work/research/passkey/#cve-2024-9956
  - https://www.inovex.de/de/blog/phishing-for-passkeys-an-analysis-of-webauthn-and-ctap/
  - https://denniskniep.github.io/posts/14-fido-cross-device-phishing/

Passkey phising



mRr3b00t
@UK_Daniel_Card

Proximity and threat are important considerations.....

9:00 nachm. · 30. Dez. 2025 · **1.584** Mal angezeigt

2 8

Deine Antwort posten

Antworten

mRr3b00t @UK_Daniel_Card · 30. Dez. 2025

getting punched or stabbed required about 1m proximity.

Bluetooth needs 10m
BLE is 100m

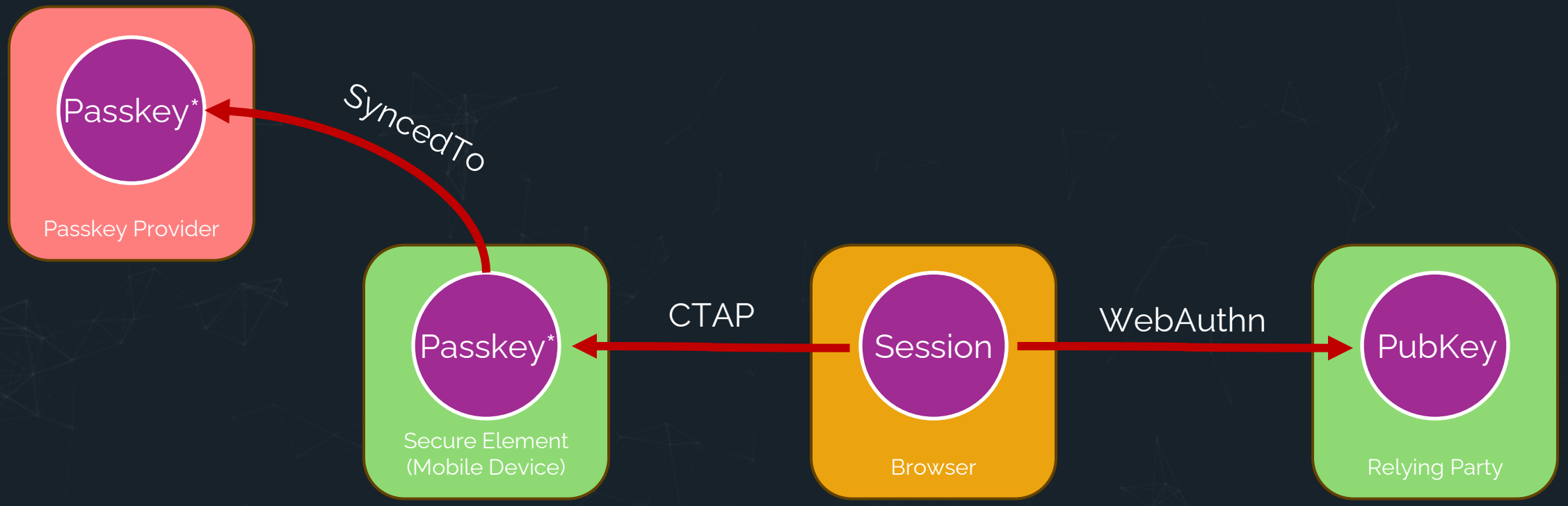
WIFI is about 40m (but really about 20m)

phishing someone can be done from any point in the world....

attacking someones web app/network kit can be done from anywhere on the planet

3 1 4 579

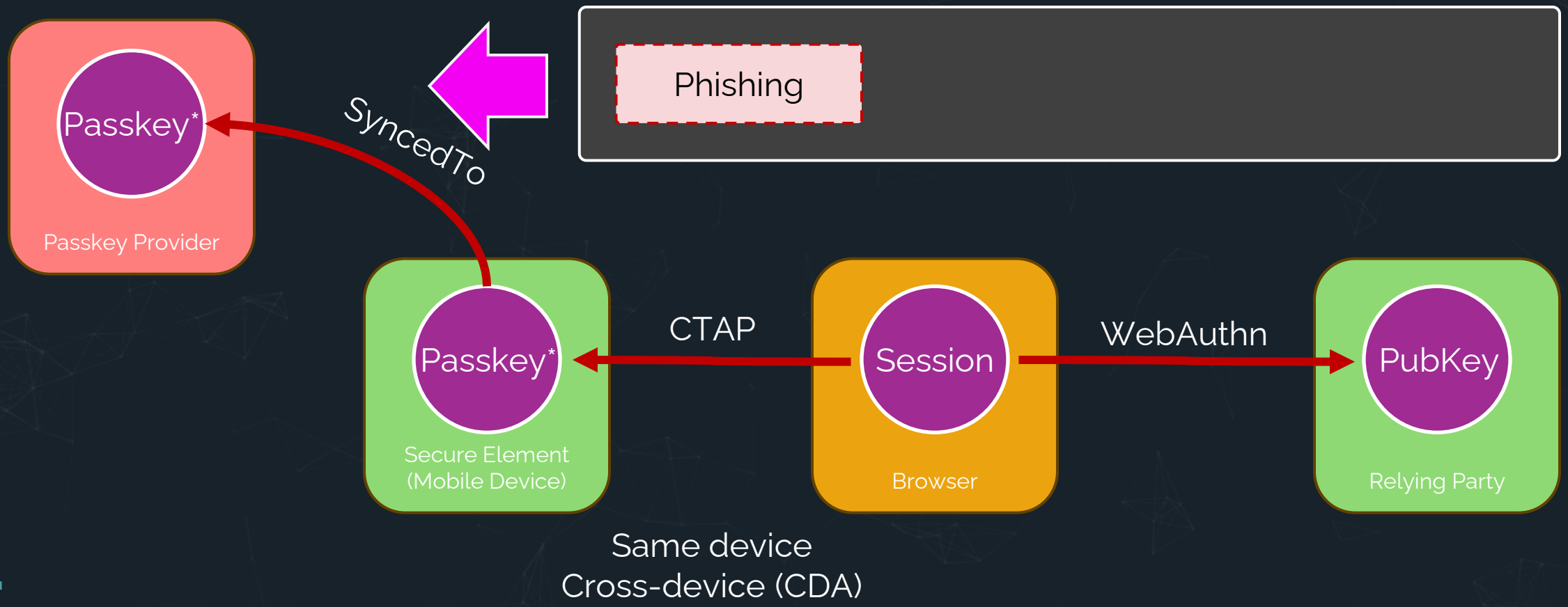
Attack vectors: Syncable passkeys



Same device
Cross-device (CDA)

*Passkey = Private Key

Attack vectors: Syncable passkeys



*Passkey = Private Key

Attacks on syncable passkeys

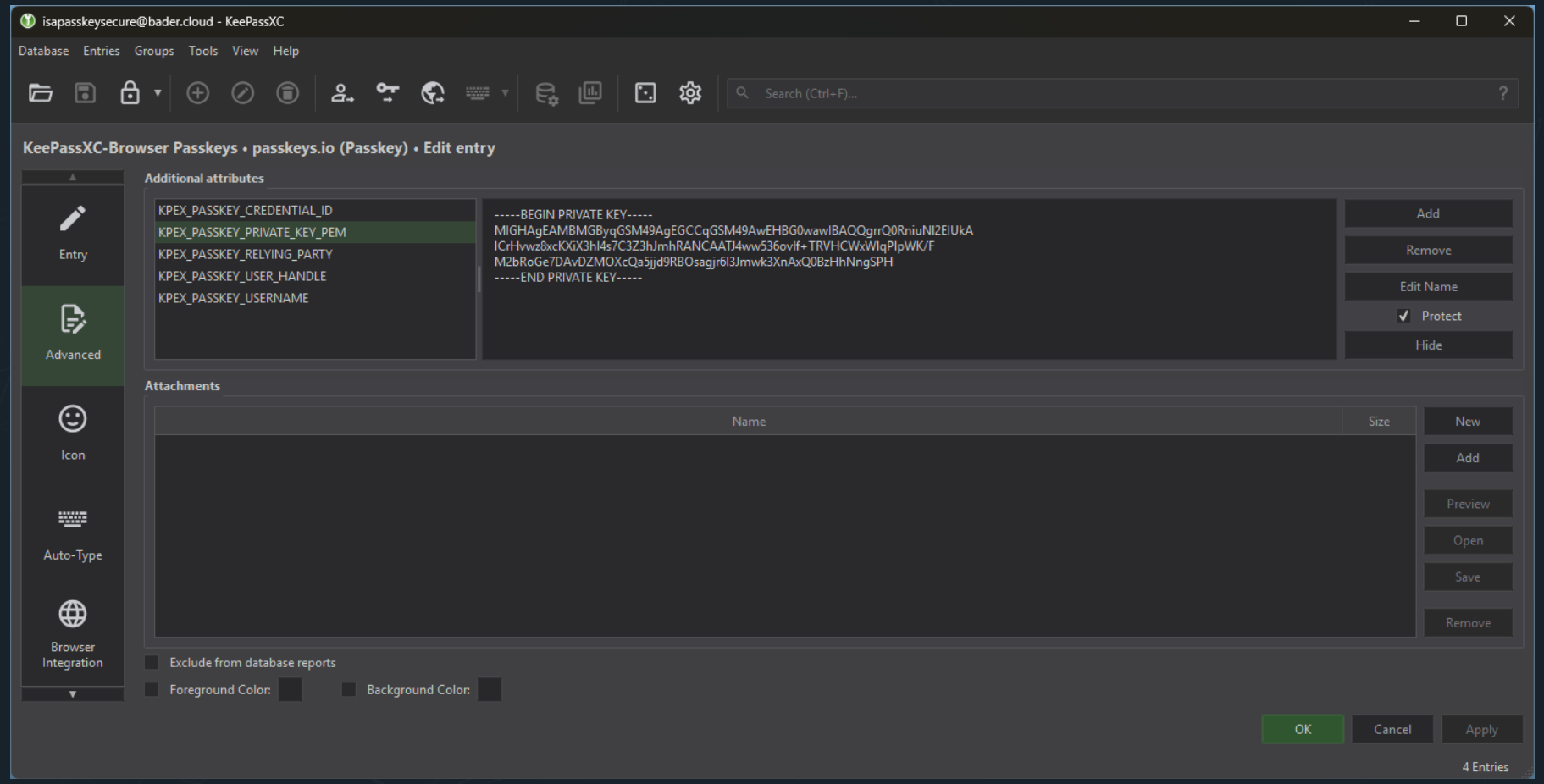
- Apple
  - Recovery: iCloud Account, Password, SMS + Device Passcode
  - 10 attempts until locked
  - https://support.apple.com/en-us/102195
- Google
  - Recovery: Google Account, Password + Device Passcode or PIN
  - https://security.googleblog.com/2022/10/SecurityofPasskeysinthe GooglePasswordManager.html
- Third parties
  - It's the wild west

Phishing of main account

Provider	MFA Required for Passkey creation	Export/ Import	New device verification	New device notification
Bitwarden	No	Yes/Yes	E-Mail	Yes
ProtonPass	No	Yes/Yes	No	No
KeePassXC	No	Yes/Yes	No	No
Keeper	No	Yes/Yes	E-Mail*	No

*Based on source IP address

Phishing of main account

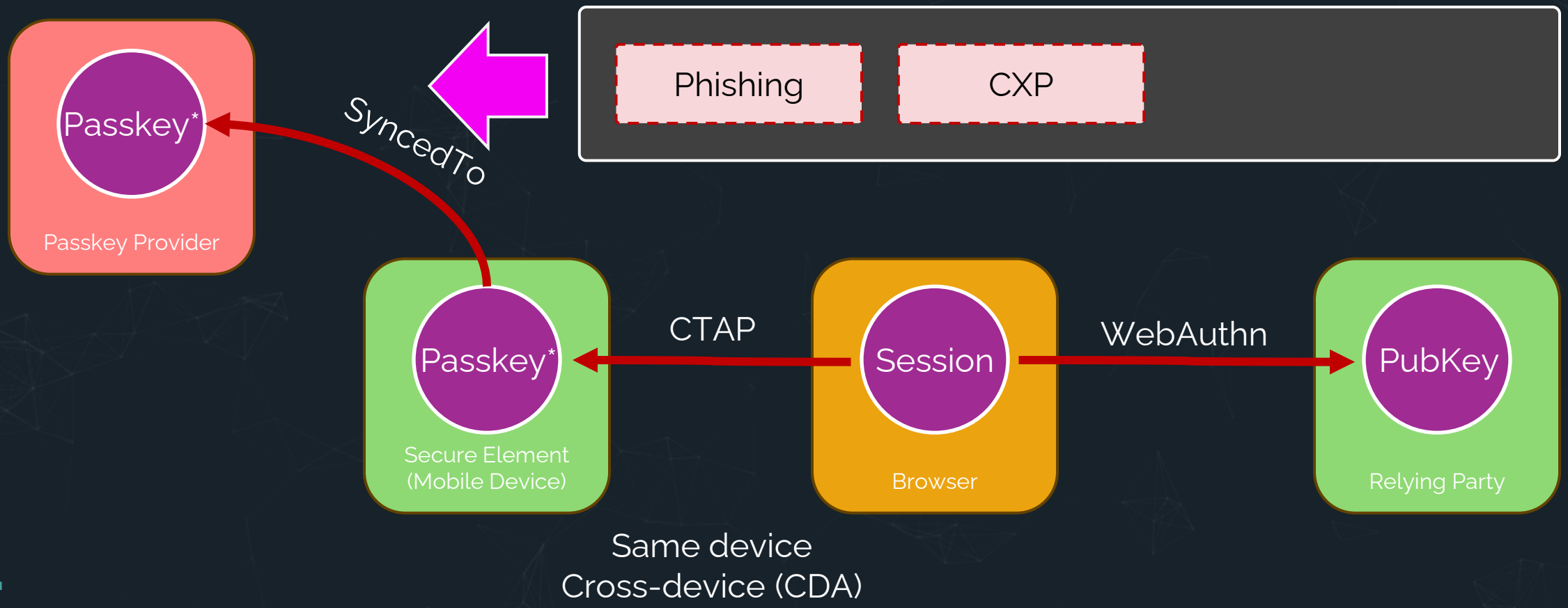


(Ab)use of exported passkeys

```
TokenTactics 0.2.21

PS C:\Users\Fabian\git\TokenTacticsV2> ipmo .\TokenTactics.psd1
PS C:\Users\Fabian\git\TokenTacticsV2> Invoke-EntraIDPasskeyLogin -KeyFilePath "C:\Users\Fabian\Downloads\testpasskeyexport\maija@c4a8korriban.com.passkey"
 -UserPrincipalName maija@c4a8korriban.com
✗ Loading key data from file: C:\Users\Fabian\Downloads\testpasskeyexport\maija@c4a8korriban.com.passkey
✔User:       maija@c4a8korriban.com
✔RP ID:      login.microsoft.com
✔Origin:     https://login.microsoft.com
✔CredID:     T1vHC2JZ9zsxXWz1hFlV6gkycRQV_gZN_CrA0It3gnU
✔UserHandle: T0Y6T2xo4yevIk-53gYvBbk6rFXfzPEAp8U36fYiav71Xh-5E7_wmxsc5MLX-foTZoNg
✗ Warming up session on login.microsoftonline.com (Authorize)...
✗ Validate FIDO2 Credential Type...
✔Challenge Received.
✗ Generating FIDO Assertion locally...
✗ Get required pre-information from microsoft.com...
✗ Submitting FIDO2 assertion to microsoftonline.com ...
✗ Submitting FIDO2 assertion to microsoftonline.com with sso_reload=true ...
✗ PageID: CmsiInterrupt
✗  Correlation Id: edda53ad-32a8-42ed-bf7e-ff281603bfca
✗  Session Id: 51e9f280-25d4-42b2-8361-7d4200f02c00
✗  Username: maija@c4a8korriban.com
✗  AADSTS50199: CmsiInterrupt
   For security reasons, user confirmation is required for this application: Microsoft Azure CLI.
✗  urlPost URL: /appverify
✗ Submitting CMSI response to microsoftonline.com ...
✔Login Successful!
⚿ESTSAUTH Cookie: 1.AXkAT2xo4yevIk-53g... saved as $global:ESTSAUTH
⚿Session saved as $global:webSession for reuse in other functions.
PS C:\Users\Fabian\git\TokenTacticsV2>
```

Attack vectors: Syncable passkeys



*Passkey = Private Key

Credential eXchange Protocol

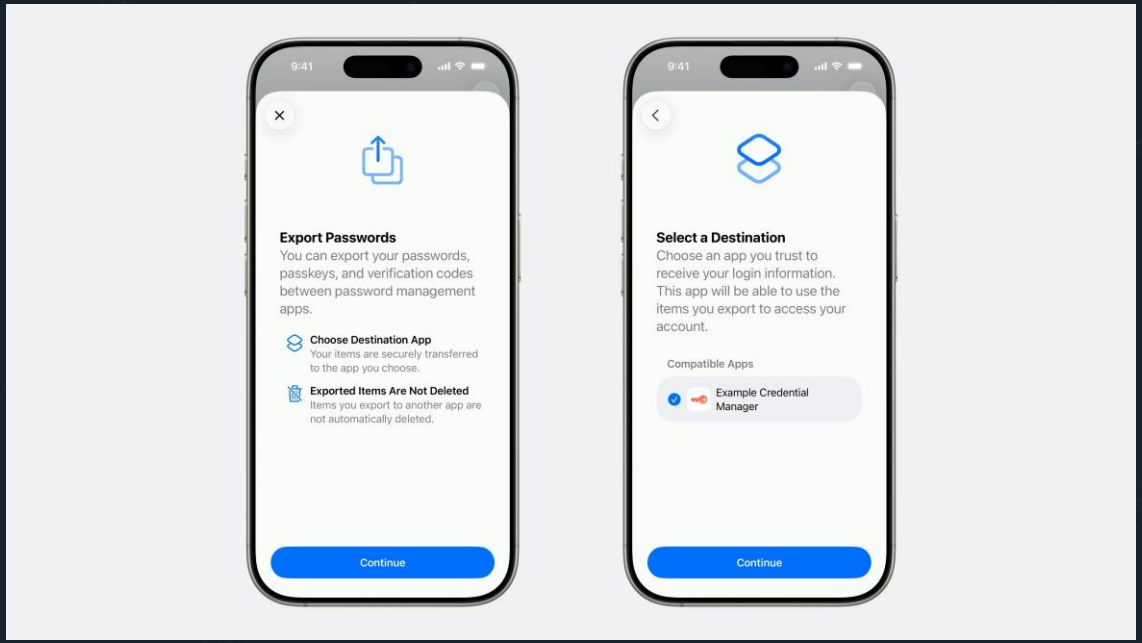
- Official method to transfer passkeys between providers
- Currently in draft
- Implemented in latest iOS versions

§ 6. Security Considerations

TODO Security

Credential eXchange Protocol

- Safe transfer of Passkeys on Apple devices
- Unsafe export from target credential manager possible



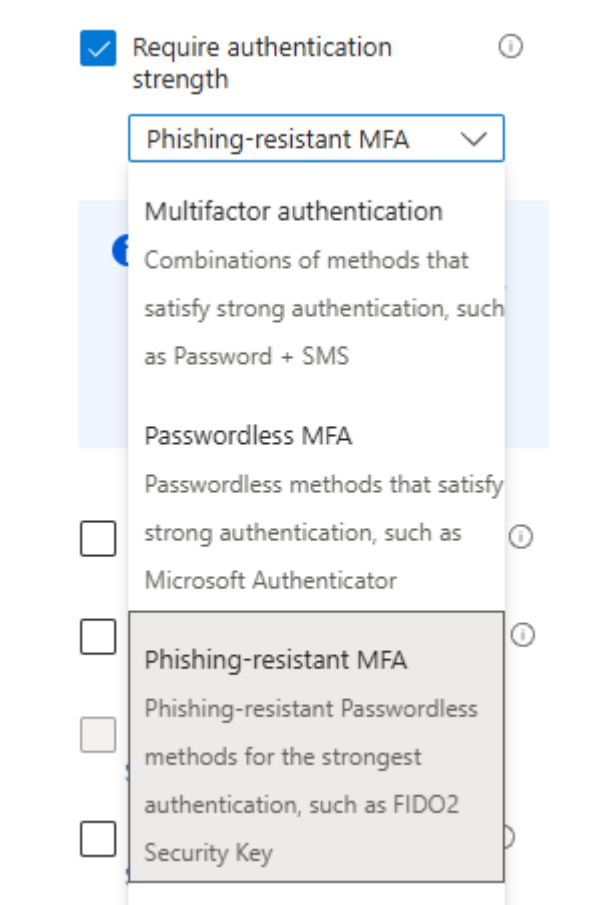
Source: Apple

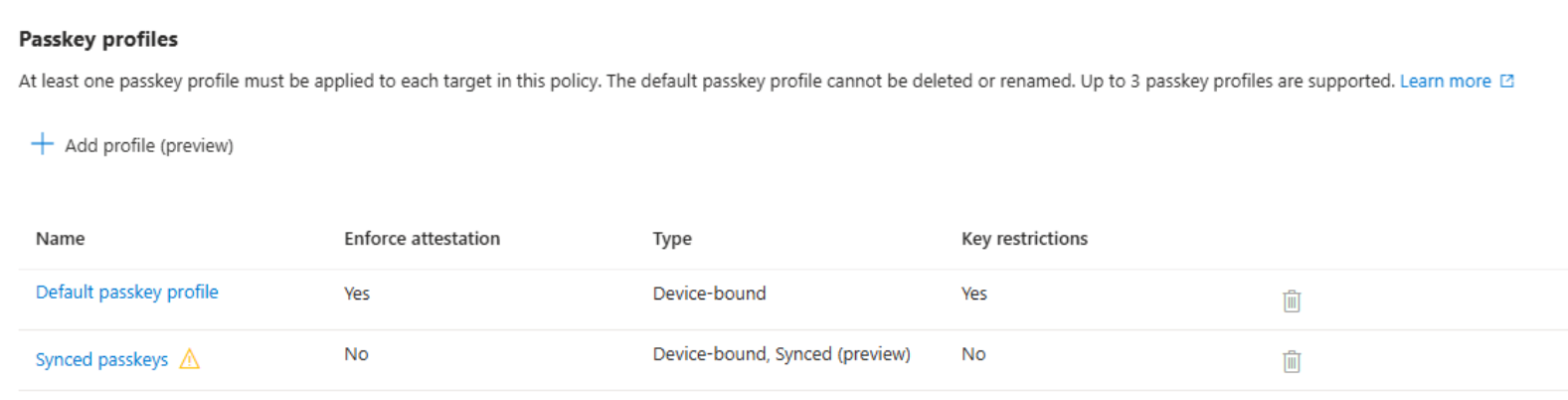
Mitigations

Mitigations in Entra ID

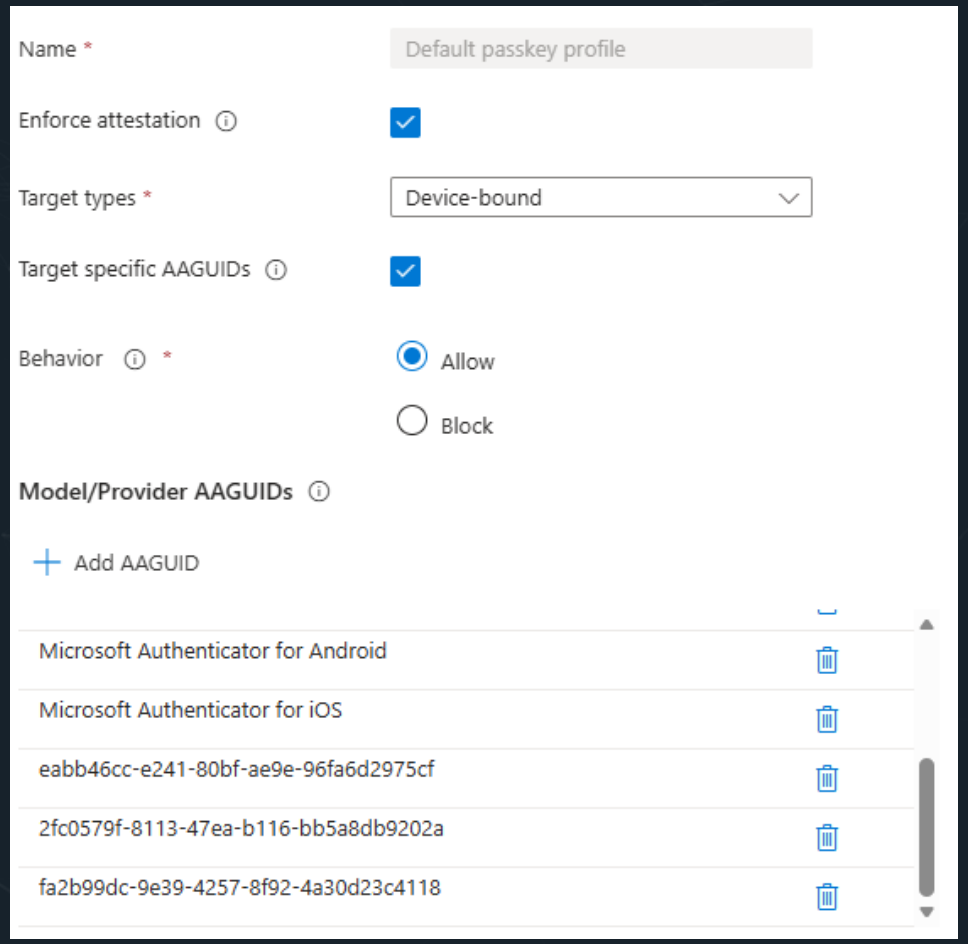
- Enforce Conditional Access Authentication Strength to only allow Passkeys
  - Recovery: Allow Temporary Access Pass from trusted location
- Use Passkey profiles to target specific audiences
  - Device-bound passkey for Administrators and targeted accounts
  - Syncable passkey for others
- Enforce attestation for device bound passkeys
- Limit device bound passkeys to verified AAGUIDs

Mitigations in Entra ID





Mitigations in Entra ID

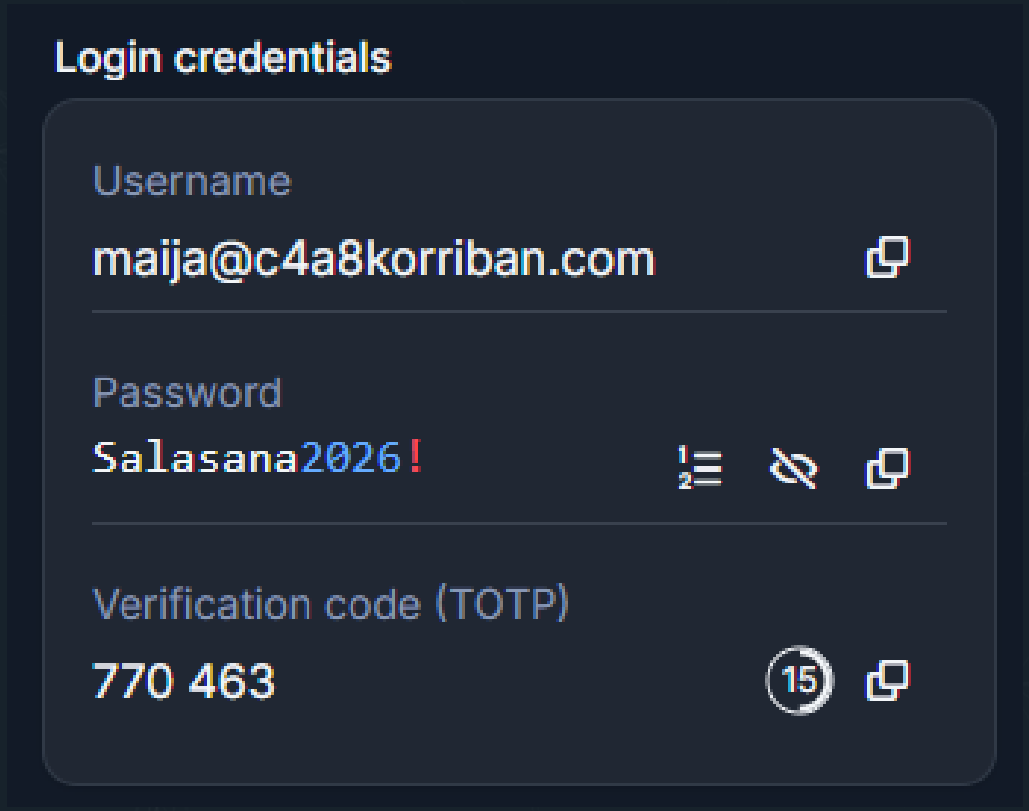


User mitigations

- Apple Advanced Data Protection
  - Fully end-to-end encryption of all data
  - Generated recovery key for own safe keeping
  - Stolen Device Protection forces biometrics outside of trusted places
- Google Advanced Protection Program
  - Requires >= 2 security key or passkey for sign-in
  - "Extra steps" for account recovery
  - Identity Check forces biometrics outside of trusted places

Conclusion

It's still 100x better than this...



Further information

- Your Passkey is Weak: Phishing the Unphishable - Chad Spensky, Ph D https://www.youtube.com/watch?v=xdl08cPDgtE
- Passkeys Pwned:Turning WebAuthn Against Itself - S Pratap Singh, J Lin, D Seetoh https://www.youtube.com/watch?v=GG4gAhbhPH8
- Google on attestation https://groups.google.com/a/fidoalliance.org/g/fido-dev/c/nhpxExcofb8/m/pd_SAJsnAwAJ
- https://www.corbado.com/blog/android-16-passkeys

Further information

- Manage passkeys in Chrome https://support.google.com/chrome/answer/13168025
- Entra ID: Enable passkeys (FIDO2) for your organization https://learn.microsoft.com/entra/identity/authentication/how-to-enable-passkey-fido2
- Assign a passkey or security key in the AWS Management Console https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_mfa_enable_fido.html

Further information

- https://github.com/f-bader/TokenTacticsV2/
- Passkeys: they're not perfect but they're getting better https://www.ncsc.gov.uk/blog-post/passkeys-not-perfect-getting-better
- Apple WWDC 2025 - What's new in passkeys https://developer.apple.com/videos/play/wwdc2025/279/?time=1152

Thank You

SwissMicrosoft
SECURITY SUMMIT

scopewyse
we are what's next

Ontinue

GRABX

INGRAM MICRO

water
IT Security & Defense

Swiss Post
Cybersecurity

techtask
Strategy · Sourcing · Security

Microsoft

SCE

IC CONSULT